

BANK OF THAILAND

Monetary Policy Report

December 2015

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
December 2015

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Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
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Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim of ensuring that monetary policy decisions are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the monetary policy target for 2015, in place of the quarterly average core inflation target of 0.5 – 3.0 percent. The new inflation target was jointly proposed by the MPC and the Minister of Finance. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the public, together with measures taken and estimated time to bring inflation back to the target.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy in 2015 is projected to expand by slightly faster than the previous assessment on the back of higher-than-expected growth in public investment and private consumption, which offset the weaker-than-expected exports and private investment in 2015 Q3. For 2016, the Thai economy is expected to grow at a pace slightly slower than previously assessed. This is due to a much larger slowdown in exports than anticipated following the slow recovery in Thailand's trading partners. However, steady growth in public spending and private consumption is projected to continue, and that should partly counter impacts from contraction in exports.

Inflationary pressures declined mainly because of supply-side factors. Headline inflation is expected to remain negative throughout 2015, but it is likely to increase gradually and turn positive in the first half of 2016, as the impact of base effects from the previously high oil prices wear out and as the economy gains traction. Core inflation is likely to be positive throughout the forecast horizon. The public's expectations of future inflation currently remain close to the inflation target.

The MPC kept the policy interest rate unchanged at the November and December 2015 meetings, judging that the overall monetary conditions remained sufficiently accommodative to support economic recovery. This was reflected in a greater amount of funds raised by the corporate sector, the real policy rate that remained at a low level, and exchange rate developments that remain supportive to the adjustment of the export sector. The Committee also gave consideration to maintaining financial stability amid the prospect of heightened volatility in the global financial markets, and to mitigating long-run risks to financial stability under the prolonged low interest rate environment.

1.1 Growth and inflation prospects

The MPC expected the Thai economy to expand by 2.8 percent in 2015, slightly faster than the previous forecast. First, higher-than-anticipated growth in public investment and private consumption partly offset weaker-than-expected exports and private investment. Second, additional government stimulus measures should provide some support to the economy for the remainder of 2015 due to stronger private sector confidence. Third, tourism, following the bombing incident in Bangkok, is likely to recover at a faster pace than the previous assessment.

Nevertheless, the Thai economy in 2016 is projected to grow slightly slower than the assessment in the previous issue of the Monetary Policy Report. Despite the better assessment of the domestic economy on the back of stronger public investment, private consumption, and tourism, it is unlikely to fully compensate for the negative impacts of weaker external demand on exports and private investment. **The Committee thus revised down the GDP growth forecast for 2016 from 3.7 to 3.5 percent (Table 1.1).**

Table 1.1 Forecast summary

Percent	2014*	2015	2016
GDP Growth	0.9	2.8	3.5
		(2.7)	(3.7)
Headline inflation	1.9	-0.9	0.8
		(-0.9)	(1.2)
Core inflation	1.6	1.1	0.9
		(1.0)	(0.8)

Note: * Outturn
( ) MPR September 2015
Source: Office of National Economic and Social Development Board, Ministry of Commerce and calculations by the Bank of Thailand

Supply-side inflationary pressures turned out lower than the earlier assessment throughout the forecast period due to a decline in domestic fuel prices in line with falling crude oil prices. Moreover, crude prices in the coming quarters are expected to increase at lower rates than our previous assumptions. Nevertheless, **demand-pull inflationary pressures is likely to rise** due to better-than-expected domestic household spending, allowing businesses to raise prices of goods and services to some degree. In addition, a planned increase in the excise tax on

automobiles is set to take effect in early 2016. As a result of these factors, the projection of core inflation has been revised up from the previous assessment, at 1.1 percent in 2015 and 0.9 percent in 2016. The Committee maintained its projection for headline inflation at -0.9 percent in 2015, while revising down the headline inflation forecast in 2016 to 0.8 percent.

The MPC assessed risks of deflation to be limited. The negative headline inflation figures could be accounted for by supply-side factors, especially falling oil prices. Prices of most of the other goods, however, continue to increase overall as reflected in core inflation remaining positive throughout the forecast period. In addition, inflation expectations remained close to the inflation target.

The Committee factored the following developments into the growth and inflation forecasts.

(1) Public spending, especially investment expenditure, and government measures aimed at stimulating private consumption have provided support to the economy.

The MPC revised up assumptions of public spending mainly on the back of increased public investment. Central government investment spending picked up as the government expeditiously committed to various projects, resulting in more effective disbursements during fiscal year 2016. Funds of extra-budgetary spending under the second phase of the economic stimulus package[1/] were also disbursed sooner than expected. Furthermore, it became clear that investment under

[1/] Infrastructure investment on water resource management and urgent road transport system development

the third phase of the economic stimulus package[2/] would be financed through the central government budget, with a large proportion of funds expected to be disbursed in early 2016. State enterprise investment should pick up in 2016 thanks to outlays for the Orange Line of the Mass Rapid Transit (MRT). This project should start earlier than previously expected given that the Cabinet approved it in principle[3/]. However, current government expenditure was revised down slightly on the back of a reduction in government transfers or unnecessary subsidies.

Looking ahead, the MPC concurred that sustained public spending would continue to underpin business sentiment and boost private investment in connection with government projects, such as investment under the public-private partnership (PPP) scheme for MRT projects, and housing development projects along the MRT lines. Additional stimulus measures aimed at boosting spending of the private sector—in particular, small and medium enterprises (SMEs) and low-income households[4/]—should spur household consumption

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[2/] The third phase stimulus package comprises (1) measures to promote living standards at the sub-district level, (2) measures to promote small-scale public investment nationwide in late-2015 that were in effect additional public investment spending, and (3) measures to promote living standards at the village level in the form of financial assistance to the private sector.

[3/] The Cabinet's decision on December 8, 2015.

[4/] Examples include (1) financial and fiscal measures (urgent phase) to promote SMEs, including low-interest credit and corporate tax cut for SMEs, (2) financial and fiscal measures to boost the real estate sector through the reduction of transfer and mortgage fees, together with mortgage provided by the Government Housing Bank for low-income property buyers, and (3) tax measures to promote domestic investment by allowing business operators to use new investment expenses for tax deduction.

and provide more liquidity for businesses, especially in the beginning of 2016 when most of the measures become effective, thereby sustaining the economic growth momentum to some extent.

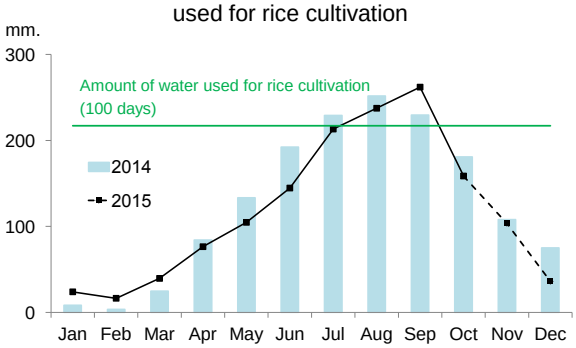
According to the Committee's assessment, overall public spending could turn out higher than the baseline assumption. The upside risk factors are that the government could accelerate investment spending more than expected—both budgetary and extra-budgetary funds—and that state enterprises could expedite new investment. The downside risk is that limited efficiency of the disbursement of public agencies could cause delays in some disbursements of funds.

(2) The impacts of drought in 2015 were in line with the previous assessment; however, water supply in the dams remained low.

The MPC evaluated the impacts of the 2015 drought to be in line with the previous assessment, according to which the amount of rainfall during May–June 2015 (which was less than usual due to the delay of the rainy season) would return to normal in 2015 Q3 **(Chart 1.1)**. As a result, farm workers increasingly moved back to the agricultural sector, having earlier moved to non-agricultural sector, especially construction and trading sectors **(Chart 1.2)**.

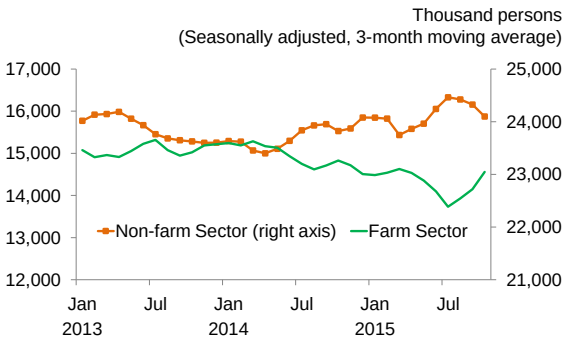
The Committee expected the drought in 2015 would cause water supply in dams to remain low through the first half of 2016. This will affect farm output, especially off-season rice production which requires irrigated water. However, based on rainfall forecasts made by several agencies predicting irregular rainfall patterns to cease in the first half of 2016, the MPC expected farm activities to return to normal in the second half of 2016, with downside risks in the case that the impacts of El Niño turn out

Chart 1.1 Ratio of amount of rainfall to amount of water used for rice cultivation



Source: Thai Meteorological Department, Royal Irrigation Department, GISTDA, calculations by the Bank of Thailand

Chart 1.2 Employment



Source: National Statistical Office, calculations by Bank of Thailand

more severe than expected. The drought should not affect the manufacturing sector because the government and businesses had made preparations, including the use of groundwater as a supplement, in the event of water shortage. Compared with the assessment in the previous issue of the Monetary Policy Report, the overall impacts of the drought have not increased and therefore should not further weigh on household consumption *(Box: Drought risks in 2016)*.

(3) Trading partners' economies are set to expand at a slower rate than previously assessed, especially with weaker-than-projected growth in Asian economies and the U.S. (Table 1.2).

The MPC revised down the assumptions of growth in trading partners' economies throughout the forecast period. In particular, Asian economies would be affected by the more pronounced slowdown in China and by the slowdown in the U.S. due to a larger-than-expected decline in exports in line with the slowdown in the trading partners of the U.S.

The slowdown in Thailand's trading partners' economies had a direct impact on the demand for Thai goods and was a factor in keeping prices of Thai exports in several categories at low levels. The slowdown in China in particular drove down commodity prices because China is a major consumer of commodities. Moreover, plunging global oil prices weighed down prices of oil-related products, such as petroleum and petrochemical products, chemicals, and rubber. Therefore, growth of Thai export value was revised down from the previous assessment both in terms of quantity and price.

Table 1.2 Growth assumptions for Thailand's trading partners

Annual percentage change (%YoY)	Weight (%)	2014	2015		2016	
			Sep 2015	Dec 2015	Sep 2016	Dec 2016
The U.S.	14.9	2.4	2.5	2.5	2.8	2.6
The euro area	10.0	0.9	1.3	1.5	1.5	1.4
Japan	13.6	-0.1	0.8	0.7	1.2	1.1
China	15.7	7.4	6.8	6.9	6.6	6.5
Asia (excluding Japan and China)*	37.4	4.2	3.7	3.5	4.2	3.9
Total**	100	3.5	3.3	3.2	3.6	3.4

Note: * Weighted by each trading partner's share in Thailand's total exports in 2014 (7 countries/economies: Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), Republic of Korea (2.8%) and the Philippines (3.7%))

** Weighted by each trading partner's share in Thailand's total exports in 2014 (13 countries)

The delay in the prospective recovery of trading partners' economies weakened private sector confidence. At the same time, declining manufacturing activities following falling exports resulted in considerable excess production capacity. Most businesses thus deferred their investment plans while awaiting clearer signs of economic recovery.

Exports of services were likely to register a steady growth. The number of foreign tourists picked up faster than expected following the bombing incident in August 2015. In particular, the number of Chinese tourists quickly rebounded thanks to marketing campaigns aimed at Chinese visitors in order to offset the impacts of a decline in the number of tourists from other countries. The downgrades of Thailand's civil aviation safety rating by the U.S. Federal Aviation Administration (FAA) is expected to have limited impacts on tourists' decision to visit Thailand, as there are relatively not many serviced flights between Thailand and the U.S. provided by Thai carriers. Moreover, the Committee was of the view that Thai tourism would not be significantly affected by the geopolitical risks provided that the problem did not persist or escalate. The Committee therefore increased the projection of the number of foreign tourists from our earlier assessment, from 29.4 million to 29.8 million for 2015, and from 30.3 million to 30.9 million for 2016.

The Fed's decision to raise the policy rate at the FOMC meeting on December 16, 2015 was consistent with the MPC's earlier assessment. The federal funds rate was expected to rise gradually so as not to undermine the recovery of the U.S. economy. However, the Fed's monetary policy normalization may create volatility in financial markets and capital flows going forward.

According to the Committee's assessment of the balance of risks with respect to the trading partners' economies, downside risks would dominate for the following reasons: (1) growth in China and Asian economies could be lower than expected; (2) the impacts of the Fed's policy rate hike on emerging market economies may be larger than anticipated; and (3) conflicts in international politics may persist and affect trading partners' economies more than the baseline projection. However, trading partners' economies may experience higher growth than the baseline projection if additional monetary easing spurred stronger growth in the euro area and Japan.

(4) Global oil prices declined by more than expected in 2015 Q4 (Chart 1.3).

The MPC revised down the assumptions for crude oil prices throughout the forecast horizon, with the average prices of 51.2 and 43.0 U.S. dollars a barrel in 2015 and 2016, respectively. The revision followed the decline in oil prices in 2015 Q4, brought about by large global oil inventory and OPEC's decision not to impose production quotas at its December 2015 meeting, which underpinned investor expectations of subdued global oil prices going forward.

The Committee thus projected crude oil prices to remain at low levels in the downward cycle for some time. The oil prices should show clear signs of upward trend again only if the global economy show clearer signs of recovery, or if there were incidents affecting the world's major oil producers **(Chart 1.4)**.

The above projection of declining global oil prices would consequently affect the outlook for domestic retail oil prices and headline inflation. Nonetheless, the Committee maintained the

Chart 1.3 Assumptions on Dubai oil price

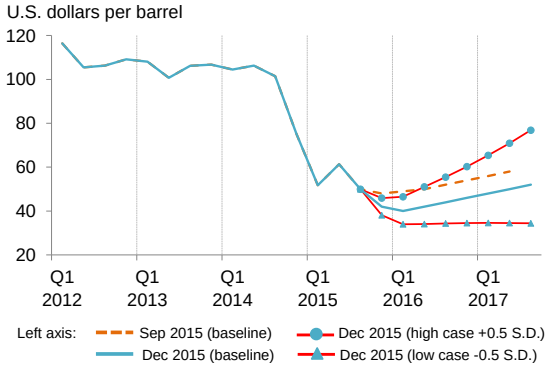
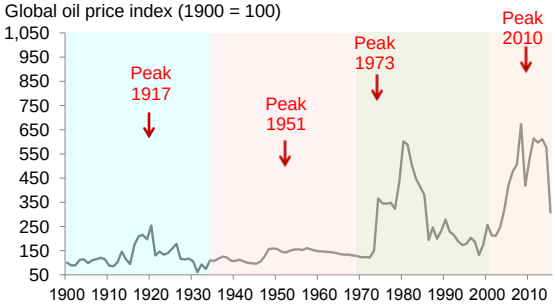


Chart 1.4 Global oil price cycle



Source: Working Paper titled "FROM BOOM TO BUST: A TYPOLOGY OF REAL COMMODITY PRICES IN THE LONG RUN" by David S. Jacks, using data from the Global Financial Data and the IMF

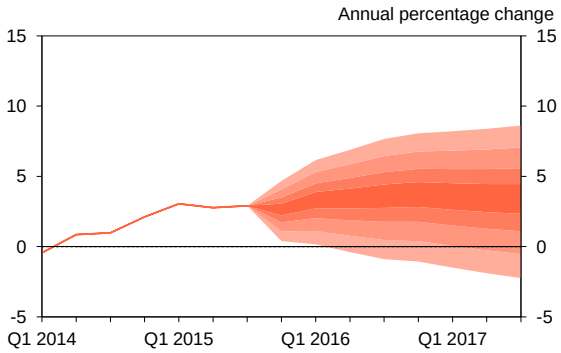
projection that headline inflation will turn positive in the first half of 2016, given the waning impacts of the high oil price base, the baseline assumption of a gradual rise in global oil prices, and the recovery in the domestic economy.

The MPC assessed the risk that the Dubai oil price would be below the baseline assumption roughly equal to the risk that it would exceed the baseline. Crude prices could fall below the baseline projection for the following reasons: (1) global demand could be lower than the current assessment if the global economy recovered only slowly; and (2) shale oil producers could expand production faster than expected. On the other hand, crude oil price could rise above baseline assumption if the Middle East conflict escalated and led to a fall in global oil production.

Risks to Growth and Inflation Forecasts

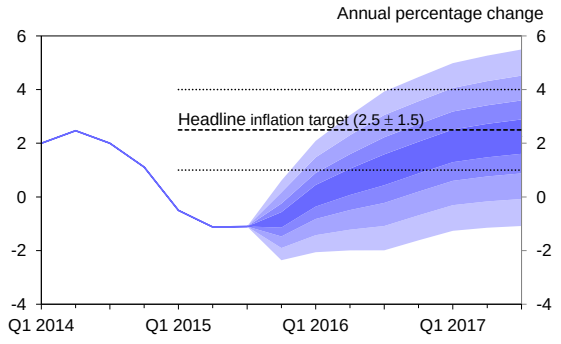
The MPC assessed risks to growth tilted to the downside, as reflected in the growth fan chart that (Chart 1.5). Downside risks to economic growth are as follows: (1) trading partners' economies, especially China and Asia, could grow more slowly than anticipated; (2) international conflicts could intensify, undermining global economic recovery and tourist confidence more than expected; and (3) drought could turn out to be more severe, affecting farm and manufacturing production by more than anticipated. On the other hand, upside risk factors to growth could arise from the following sources: (1) the government's economic stimulus measures could have a more pronounced impact on private spending than expected; and (2) public spending, especially on investment projects, could be expedited more quickly than anticipated. Overall, the above risks could cause growth to be higher or lower than the

Chart 1.5 GDP growth forecast



Note: The fan chart covers 90 percent of the probability distribution.

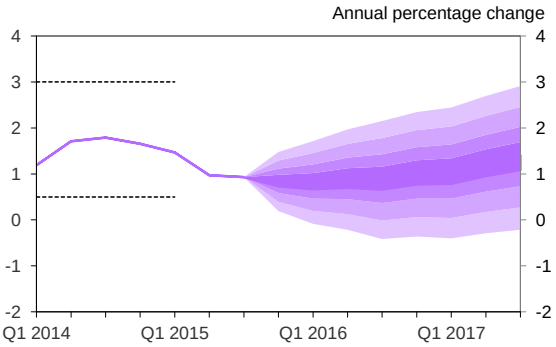
Chart 1.6 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

baseline projection, depending particularly on the impact of uncertainty on business investment. For **headline and core inflation, the Committee judged the balance of risks also tilted to the downside (Charts 1.6 and 1.7)** based on the risk that demand-side pressures could be softer if economic growth happened to fall short of the baseline projection.

Chart 1.7 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

1.2 Monetary policy decision

The MPC kept the policy rate unchanged at its November 4 and December 16, 2015 meetings, judging that overall monetary policy conditions remained sufficiently accommodative to support economic recovery. This was reflected in the increase in the amount of funding raised by the business sector, the real policy rate that remained steady at a low level, and exchange rate developments that remain supportive to the adjustment of the export sector. The Committee also gave consideration both to maintaining financial stability under the prospect of heightened global financial market volatility and to mitigating risks to long-term domestic financial stability that could accumulate under a prolonged low interest rate environment.

Looking ahead, the Committee was of the view that monetary policy stance should continue to be accommodative to support Thailand's economic recovery. The Committee would continue to closely monitor developments in the global financial markets and the impacts of the domestic fiscal stimulus on economic recovery. In addition, the MPC would monitor and evaluate imbalances which might build up, such as search-for-yield

behavior and any impacts on longer-term saving, as a result of the prolonged low interest environment.

At the MPC meetings on November 4 and December 16, 2015, the Committee assessed Thailand's economic recovery to be gradual, driven mainly by public spending and tourism growth. Moreover, overall private spending was projected to expand during the last quarter of the year. However, economic recovery faced a number of negative factors due to stronger evidence of a slowdown in major trading partners' economies, especially in China and the ASEAN. Overall inflationary pressures stabilized at low levels due primarily to declining global oil prices. Going forward, as the base effect starts to disappear and with economic recovery, headline inflation was expected to increase gradually and turn positive in the first half of 2016, which was pushed back from our previous projection.

Moreover, global and regional financial market conditions remained volatile, mainly because of shifting market expectations over the timing of the Fed's policy rate hike. From the second half of September to October, global financial market volatility temporarily declined, with capital inflows returning to emerging Asian markets as the markets expected that the Fed would not increase the policy rate in October. This resulted in a rather rapid appreciation of the baht against the U.S. dollar in the first half of October. Between the MPC meetings on November 4 and December 16, 2015, it became more likely that the Fed would raise the policy rate in December, as the U.S. economic indicators turned to be better than market expectations. In addition, due to the Federal Open Market Committee's (FOMC) communication on monetary policy going forward, foreign investors continuously reduced their asset

holdings in emerging markets, including Thailand. This led to a return of greater financial market volatility and the baht depreciation against the U.S. dollar.

Under these circumstances, the MPC voted unanimously to leave the policy rate unchanged at 1.50 percent at the two meetings. The decision took into consideration the effectiveness of additional policy easing in facilitating economic recovery, which remained gradual and not yet broad-based, and low inflationary pressures versus the consequences of further easing on financial market stability and long-term financial stability. The MPC concluded the following. First, further policy easing having only marginal effects on the real economy partly because banks remained cautious about lending given concerns over deteriorating loan quality. Second, overall monetary conditions were deemed accommodative as reflected in the increased funding raised by the business sector and the real interest rate that remained at a low level. Third, exchange rate developments remained conducive to economic recovery despite displaying some fluctuations in some earlier periods. Fourth, government spending and economic stimulus would help shore up private sector confidence and boost business demand for activities related to public investment. Therefore, the policy rate was left unchanged at these meetings in order to preserve the policy space for downside risks that could arise in the future, for instance, the possibility that the Chinese slowdown might prove to have more negative impacts on the Thai economy than previously assessed.

The MPC agreed that monetary policy at this juncture should give more weight to preserving financial market stability by preventing

further uncertainties at the time when financial markets remained volatile and sensitive to monetary policy shifts in major industrial economies. Some MPC members were concerned with exchange rate adjustments which, at times, may not have been conducive to economic recovery. This situation, together with the impacts of the Fed's monetary policy shift on the baht, warranted constant monitoring. A majority of MPC members stressed the importance of monitoring financial stability risks, such as the search-for-yield behaviors, that could arise in the prolonged low interest rate environment.

The Committee agreed that, going forward, monetary policy should remain adequately and continually accommodative to support gradual recovery of the Thai economy. Government measures would also gradually play a greater role. However, changes in the global trade structure and Thailand's declining competitiveness in some product categories would continue to weigh on the recovery of the export sector despite the continued recovery in the global economy, in particular the advanced economies. The MPC thus underlined the need for structural adjustment in the export sector, in terms of both goods and services, in order to raise competitiveness. The MPC would continue to closely monitor downside risks to the Thai economy, especially the slowdown of the global economy and uncertainty in financial markets. The Committee would stand ready to utilize all available policy tools to support economic recovery and safeguard financial stability.

1.3 Appendix: Tables for supporting assumptions and forecasts

Table 1.3 Forecasts for GDP and Components			
Percent (per annum)	2014*	2015	2016
GDP growth	0.9	2.8	3.5
Domestic demand	-0.1	2.6	3.5
Private consumption	0.6	2.4	2.8
Private investment	-2.0	-2.0	3.9
Government consumption	1.7	2.6	3.4
Public investment	-4.9	23.1	8.8
Exports of goods and services	0.0	0.0	2.1
Imports of goods and services	-5.4	-0.6	3.1
Current Account (billion US dollars)	14.2	34.0	26.3
Value of merchandise exports	-0.3	-5.5	0.0
Value of merchandise imports	-8.5	-11.4	0.9

Note : *Outturn

Table 1.4 Forecast assumptions			
Annual percentage change	2014	2015	2016
Dubai oil price (U.S. dollars per barrel)	96.7	51.2	43.0
Non-fuel commodity prices (%YoY)	-3.9	-17.4	-3.0
Fresh food prices (%YoY)	4.8	-3.9	3.2
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY)[1/]	3.4	4.4	7.5
Public investment (current price) (%YoY)[1/]	-4.7	20.2	12.2
Fed Funds rate (% at year-end)	0.13	0.38	1.38
Trading partners' economic growth (%YoY)[2/]	3.5	3.2	3.4
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	135.9	151.4	157.6

Note : [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by the minus sign

Table 1.5 GDP growth forecasts by research houses		
	2015	2016
Maybank Kim Eng	3.0	4.5
Standard Chartered	2.9	4.0
NESDB[1/]	2.9	3.5
Bank of Ayudhya	2.9	3.2
FPO[2/]	2.8	3.8
BOT	2.8	3.5
Kasikorn Research	2.8	3.0
KT Zmico	2.8	3.0
Tisco	2.7	3.5
TMB	2.7	3.5
DBS	2.7	3.4
Phatra	2.7	3.2
JP Morgan	2.7	2.9
ANZ Bank	2.6	3.5
HSBC	2.6	3.3
SCB	2.5	2.5

Note: Compiled and published by Reuters on December 14, 2015, except:
[1/] Published on November 16, 2015, with the release of GDP data for 2015 Q3
[2/] Published on October 28, 2015
Presented in descending order of 2015's forecast

Table 1.6 Headline inflation forecasts by research houses		
	2015	2016
Maybank Kim Eng	-0.5	2.5
ANZ Bank	-0.6	1.6
Standard Chartered	-0.6	1.4
HSBC	-0.8	1.6
NESDB[1/]	-0.8	1.5
KT Zmico	-0.8	1.3
Tisco	-0.8	1.3
Phatra	-0.8	1.1
FPO[2/]	-0.9	1.8
Bank of Ayudhya	-0.9	1.5
DBS	-0.9	1.5
Kasikorn Research	-0.9	1.2
Thanachart Securities	-0.9	1.1
SCB	-0.9	0.9
TMB	-0.9	0.9
BOT	-0.9	0.8
JP Morgan	-0.9	0.6

Note: Compiled and published by Reuters on December 14, 2015, except:
[1/] Published on November 16, 2015 with the release of GDP data for 2015 Q3
[2/] Published on October 28, 2015,
Presented in descending order of 2015's forecast

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2015	2016				2017		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
10-12	0	0	0	0	1	1	1	2
8-10	0	0	1	3	4	5	5	5
6-8	1	10	14	18	19	18	18	17
4-6	12	23	22	20	20	18	17	16
2-4	53	41	34	29	27	26	25	24
0-2	31	21	21	19	18	19	19	19
(-2)-0	3	4	6	8	8	9	10	11
< (-2)	0	0	1	2	3	4	5	6

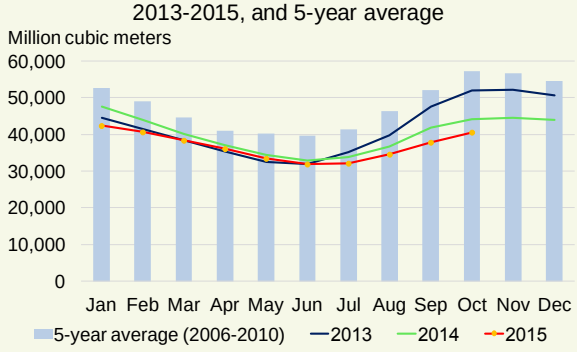
Table 1.8 Probability distribution of headline inflation forecast								
Percent	2015	2016				2017		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
6-7	0	0	0	0	1	1	2	2
5-6	0	0	0	1	2	3	4	5
4-5	0	0	1	3	6	8	9	10
3-4	0	1	4	8	12	15	16	16
2-3	0	5	12	16	18	20	20	20
1-2	2	16	21	21	21	20	19	18
0-1	15	29	26	21	19	16	15	14
(-1)-(0)	39	28	20	16	12	10	9	8
(-2)-(-1)	34	15	11	9	6	5	4	4
(-3)-(-2)	10	5	4	4	2	2	1	1
(-4)-(-3)	1	1	1	1	1	0	0	0

Table 1.9 Probability distribution of core inflation forecast								
Percent	2015	2016				2017		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
3.0-3.5	0	0	0	0	1	1	2	3
2.5-3.0	0	0	1	1	3	3	5	7
2.0-2.5	0	1	4	6	8	9	12	14
1.5-2.0	4	9	13	14	16	17	19	19
1.0-1.5	30	27	26	23	23	22	22	20
0.5-1.0	46	35	28	25	23	21	19	17
0.0-0.5	18	21	19	18	16	15	12	11
(-1)-0.0	2	6	7	9	8	8	6	5
(-2)-(-1)	0	1	2	3	3	3	2	2
< -2	0	0	0	1	1	1	1	1

Drought risks in 2016

Every year from November to April, Thailand experiences its dry season, with lower rainfall than in any other times of the year. Most water usage during this time will rely mainly on water from dams and natural reservoirs, where water supply depend on the amount of rainfall during the previous year's rainy season. However, volatile weather patterns in some years could lead to drought. During the last months of 2014, El Niño caused irregular rainfall and low water supply in dams in November 2014. The water supply is used for consumption, preservation of the ecosystem (e.g., saltwater push back), industrial purposes, and farming activities during the dry season of 2015 (between November 2014 and April 2015). As a result, the government asked farmers to refrain from planting rice in the off-season—that would require significant use of water—so as to ensure adequate water supply until the subsequent rainy season which started in May 2015. These policy measures resulted in a 43 percent reduction in off-season rice production in 2014/2015 compared with the previous crop year.

Chart 1 Monthly water levels in Thailand's dams, 2013-2015, and 5-year average



Source: Royal Irrigation Department

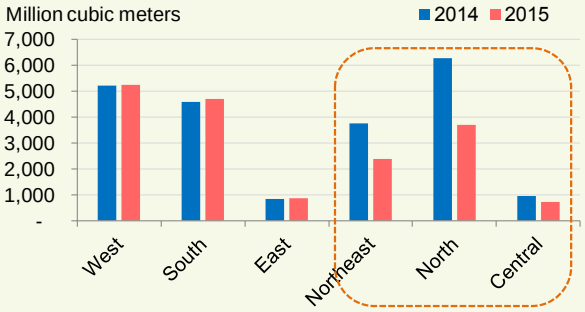
At the start of the 2015 rainy season, El Niño still persisted and later became most severe since 1997. Rainfall delayed during January–June 2015 and turned out to be lower than the previous year and indeed significantly below the historical average. As a result, the country-wide water supply in dams as of 1 November 2015 dropped 29 percent below the usual 5-year average (2006-2010) and 8 percent lower than the previous year's figure **(Chart 1)**. Concerns heightened over the possible shortage of water for economic and environmental activities. It was therefore necessary for the government to introduce additional measures to reduce in-season rice planting, causing the in-season rice production to fall from the previous year.

A closer look at the water supply in dams in various regions shows that available water supply in the Northern, Northeastern, and the Central regions is much lower than the previous year **(Chart 2)** especially in the major dams (namely, the Bhumibol, Sirikit, and Pasak Jolasid Dams. Consequently, there are risks that the available water supply may be insufficient for use during the first half of the dry season in 2016. Given the current water supply and coupled with the persistent impacts of El Niño, the drought situation may worsen. The government has therefore issued the Water Management Plan for the Drought Season in 2016. The Royal Irrigation Department plans an additional 4 percent reduction in irrigated water next year[1/] **(Chart 3)** and has prioritized water usage activities in the order of importance, namely water for consumption, preservation of the ecosystem, industrial purposes, and farming respectively. The amount of water supplied to non-farm activities is not much relative to the amount irrigated to farm activities. For the manufacturing sector, in addition to the government's water management plan, most industrial estates have also

[1/] Source: Water Management and Crop Cultivation in Drought Season in Irrigation Area Plan, 2015/2016.

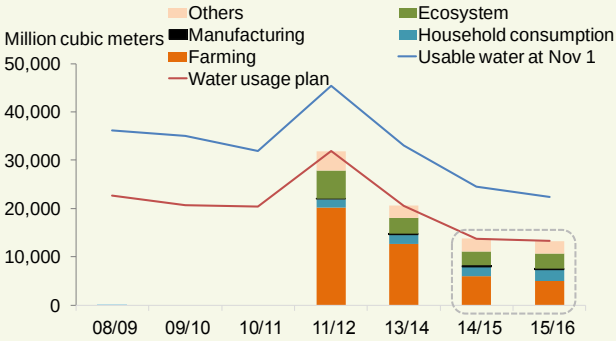
devised plans to deal with the potential water shortage in 2016. Consequently, the drought is estimated not to have significant impacts on the manufacturing sector. Nonetheless, some areas located in the downstream of the Chao Phraya and Mae Klong river basins are prone to the risk of saltwater intrusion along the streams, which degrades the quality of water and make it unsuitable for manufacturing purposes.

Chart 2 Usable dam water levels by regions, end of October 2014 and 2015



Source: Water Watch and Monitoring System for Warning Center, Royal Irrigation Department

Chart 3 Water usage by categories listed in the water management and dry season cultivation plan, 2011-2015

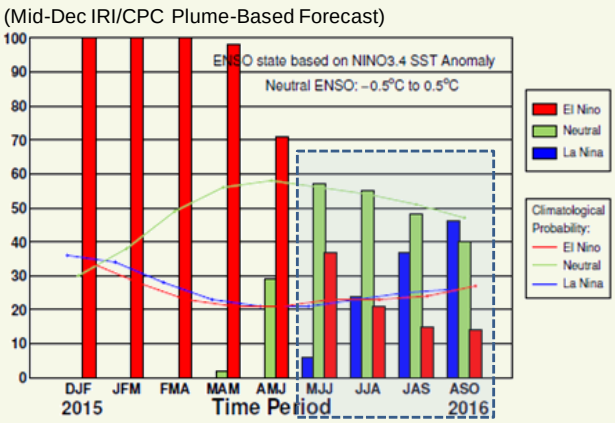


Source: Office of Water Management and Hydrology

In terms of water discharged for farming in 2016, the government plans to cut water supply by 17 percent from the previous year. This will affect off-season rice growing, which requires high water usage, especially in the Northern, Central, and Northeastern regions that altogether account for 97 percent of the country's total off-season rice farming area. Crops that need little water (namely sugarcane, cassava, and corn) may be somewhat affected. The government has recognized the potential impacts of the drought and has therefore introduced measures to assist affected farmers. These include training and providing production inputs, together with financial assistance such as loan repayment postponement and extension for farmers who have borrowed from state financial institutions.

With regard to the outlook for the rainy season in 2016 (May–October), the probability of El Niño during this period is lower compared with the first half of the year according to the International Research Institute for Climate and Society (IRI)[2/] **(Chart 4)**. However, it remains a risk that requires monitoring. The BOT expects drought to result in a contraction in off-season rice production in the first half of 2016 relative to the same period last year. Therefore, farm income growth is projected to be only 2.3 percent and could be lower if the impacts from the El Niño are greater than forecast. Incidentally, in the absence of drought, farm income is estimated to rise by 3.1 percent.

Chart 4 El Niño forecast, December 2015



Note: Alphabets on the horizontal axis are abbreviations of names of the months in English

Source: IRI, ENSO

[2/] International Research Institute for Climate and Society, Columbia University

However, drought remains a long-term problem that could undermine economic activities and significantly increase uncertainty for the farm income outlook in the future. Every stakeholder should be involved in tackling this problem. The level of general awareness should be raised. The public needs to understand the value of water resources and the increasing risks of drought in order to induce a change in behavior and promote more responsible and efficient water consumption. The government, industries, farmers, and citizens should deepen cooperation to ensure efficient water collection, allocation, and management for the benefit of everyone in the society and in preparation for the risks of drought and volatile weather patterns in the future.

Recent Economic Developments

2. Recent Economic Developments

The global economy was likely to expand at a slower pace than forecast in the previous Monetary Policy Report due mainly to the economic slowdown in China and other Asian economies. Advanced economies continued to grow: the U.S. economy posted strong growth, while the euro area and Japan were likely to pick up gradually. However, downside risks to global growth increased due to the continued slowdown of the Chinese economy, persistently depressed commodity prices, and ongoing geopolitical risks. Financial conditions in emerging markets could experience more volatility following the Fed's policy rate increase, which in effect brought about greater monetary policy divergence among major economies.

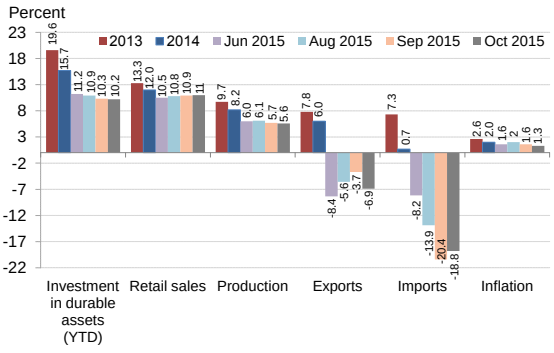
The Thai economy continued to recover, driven mainly by public spending and private consumption. Tourism began to show clear signs of recovery in October 2015. However, the recovery was still not robust as exports remained weak, weighed down by the slowdown in trading partners, especially the ASEAN economies which were increasingly affected by China's slowdown. Moreover, Thai exports were also held down by the loss of competitiveness resulting from structural problems and changes in the global trade structure. Private investment so far remained subdued.

2.1 The global economy

The global economy grew at a slower pace than assessed in the previous Monetary Policy Report, given the weaker outlook for China and Asian economies and despite the continued growth in advanced economies.

China and other Asian economies (excluding Japan) expanded at a slower rate than in previous quarters. In 2015 Q3, the Chinese economy grew by 6.9 percent (yoy), slightly slower than 7.0 percent in the second quarter. Whereas China's service sector still expanded, the manufacturing sector continued to slow. Preliminary data for 2015 Q4 pointed to a

Chart 2.1 China's economic indicators
(percentage change from the same month last year)



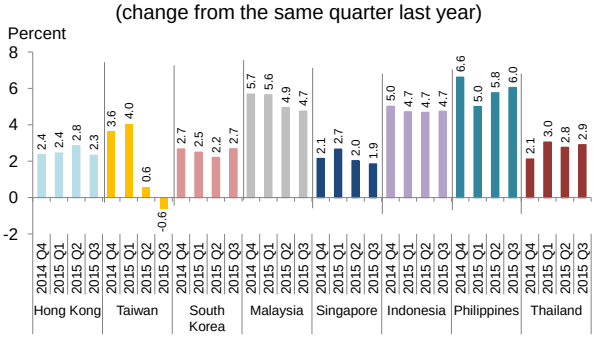
Source: CEIC

further overall slowdown owing to sluggish investment especially in the real estate sector with elevated excess supply. In addition, investment in the manufacturing sector remained subdued because firms in this sector were constrained by weaker financial positions and rising debt **(Chart 2.1)**. Nevertheless, consumption remained broadly stable as reflected in the retail sales reading broadly similar to the previous quarter, thereby somewhat supporting economic growth. In October 2015 **the Chinese administration eased monetary measures by reducing the policy interest rate and the Reserve Requirement Ratio (RRR) to provide additional economic stimulus.**

The Chinese economic slowdown, especially in investment, weighed down China's imports and global commodity prices, which adversely affected Asian exports. In 2015 Q3, several **Asian economies** (excluding China and Japan) continued to post lower growth than in the previous quarter **(Chart 2.2)**. This was due to weaker domestic demand and shrinking net exports, particularly exports to China, despite the increase in the volume of exports by some countries thanks to the launch of new electronic products. Preliminary Q4 data suggested that Asian economies (excluding China and Japan) would continue to grow at a slower pace as a result of subdued domestic demand and weakening net exports **(Chart 2.3)**.

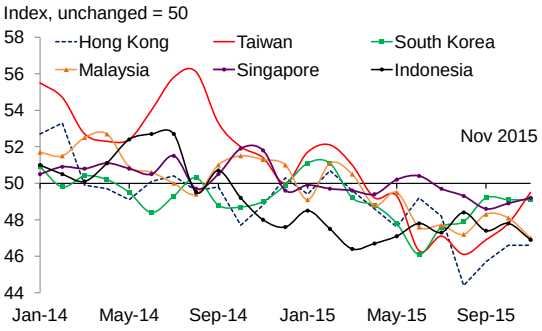
Advanced economies on the whole continued to expand (Chart 2.4). In 2015 Q3, **the U.S. economy** grew by 2.1 percent (qoq saar[1/], second estimate), down from 3.9 percent in the preceding quarter. The weaker growth was

Chart 2.2 GDP growth of Asian economies
(change from the same quarter last year)



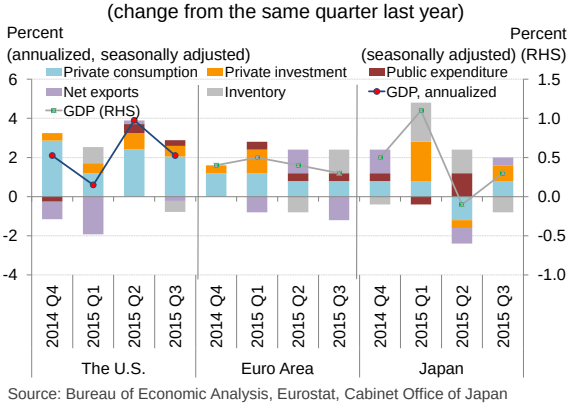
Source: CEIC

Chart 2.3 Purchasing Manager Index (PMI)



Source: Bloomberg

Chart 2.4 Sources of GDP growth of G3 economies
(change from the same quarter last year)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

[1/] Seasonally adjusted annualized rate

attributed to a decline in inventory build-up, which had taken place during the first half of the year, and to a contraction of net exports, which resulted from a stronger U.S. dollar and weak demand from trading partners' economies. Meanwhile, domestic demand was strong, supported by growing consumer confidence, higher income, and more favorable labor market conditions.

The euro area economy[2/] grew at a rate of 0.3 percent (qoq sa) in 2015 Q3, slightly down from 0.4 percent in the previous quarter. Germany and Italy posted slower growth, while France and Spain continued to expand. Growth in the euro area was underpinned by private consumption on the back of low oil prices and improving employment. Accommodative monetary policy should foster continued expansion of credit and private consumption in Q4. However, weak exports remained a factor weighing on the economy.

Japan returned to growth in 2015Q3, showing a 0.3 percent (qoq sa) expansion following a 0.2 percent contraction in the preceding quarter. The rebound was partly because of the pickup in demand after the unusually inclement weather in the previous quarter and because of growth in non-residential private investment. Preliminary Q4 data, such as consumer confidence and automobile sales, indicated a gradual improvement in private spending.

Looking ahead, the Committee judged that major trading partners would likely face slower growth than previously assessed and a more pronounced difference between advanced and emerging market economies. Downside risks increased from the earlier assessment, reflecting

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[2/] Consists of 19 countries using the euro as the single currency.

China's further slowdown, persistently low commodity prices, and geopolitical risks. These risks could affect consumer confidence and consumption in the euro area. There are also near-term downside risks from greater financial market volatility in emerging economies following the Fed's policy normalization, which reflects a monetary policy divergence among major economies.

Going forward, China is expected to expand at a slower pace due to decelerating investment following economic rebalancing and given financial and fiscal reforms. However, private consumption likely continues to grow due to strong labor market conditions and supportive policies by the government, such as urbanization reform and the 13th Five-Year Plan on National Economic and Social Development aimed at raising household income. Exports are expected to gradually improve in line with the global recovery, especially in the U.S.

Asian economies (excluding China and Japan) are projected to slow down due mainly to weaker exports amid the slow recovery of global demand. In particular, demand from China and net commodity exporters continues to be subdued (owing, in turn, to their weak exports and the significant drop in export revenues in line with depressed commodity prices). This weakened external demand contributes to subdued demand for exports from Asian economies. Moreover, Asia's recovery likely faces further downside risks to financial stability given the Fed's monetary policy normalization. Businesses in some Asian countries are highly reliant on foreign borrowing and consequently susceptible to the risk of higher costs of borrowing *(Box: Assessing corporate debt risks in emerging markets and Thailand)*.

The U.S. economy is expected to grow steadily underpinned by improved fundamentals, with downside risks from a slowdown in trading partners that could affect its exports. In addition, uncertainty remains over the impacts of the Fed's monetary policy normalization that could lead to greater volatility in capital flows or to dollar appreciation that could affect net exports, inflation, and economic recovery going forward.

The euro area is forecast to recover gradually supported by accommodative monetary policy and low oil prices. Moreover, exports should pick up as a result of a weaker euro. However, the positive impact on exports could be partially offset by a weak global demand outlook, especially in emerging market economies. Downside risks that are increasingly relevant are greater geopolitical risks following the terrorist attacks in Paris. The incident is expected to weigh on tourism and private consumption in the euro area. In this regard, the MPC will continue to follow developments relating to these geopolitical risks and their impacts on various economies.

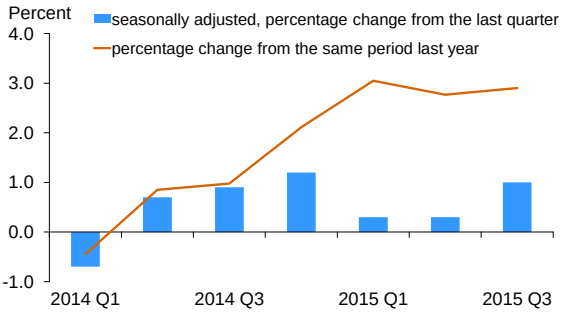
Japan is expected to recover at a slow pace thanks to the gradual expansion in private consumption and corporate profits. However, exports remain vulnerable to risks and may be constrained by slower-than-expected recovery in advanced economies and a slowdown in emerging market economies.

2.2 The domestic economy

The Thai economy continued to recover in 2015 Q3 (Chart 2.5) driven by sustained public spending. Low oil prices and growing consumer confidence bolstered growth in private consumption, especially in the necessity goods category. However, exports continued to contract.

Nevertheless, economic recovery was not yet robust as growth was limited to only some sectors. Economic indicators in October 2015 suggested a gradual recovery, driven by sustained public spending and private consumption in necessity goods and services. Tourism showed clearer signs of recovery. However, exports remained subdued in line with weak activities in trading partners' economies, especially the ASEAN countries which faced greater impacts of China's slowdown. Moreover, exports remained held down by structural problems that constrained Thailand's competitiveness and also by the shifts in global trade structure. Weaker export activity and household spending on durable goods both subdued manufacturing production and private investment.

Chart 2.5 GDP growth[1/]
(seasonally adjusted, percentage change from last quarter)

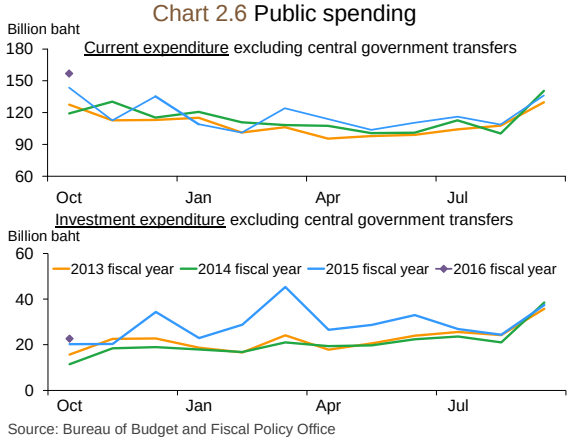


Note: [1/]Calculations based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board

Public spending continued to expand.

Public spending, both current and investment expenditures, remained a key driver of the economy. Funds for small-scale transport and irrigation investment projects continued to be disbursed. In addition, in the beginning of fiscal year 2016, budget disbursements were expedited **(Chart 2.6)** for the renovation of university buildings and structures, lending support to construction businesses.

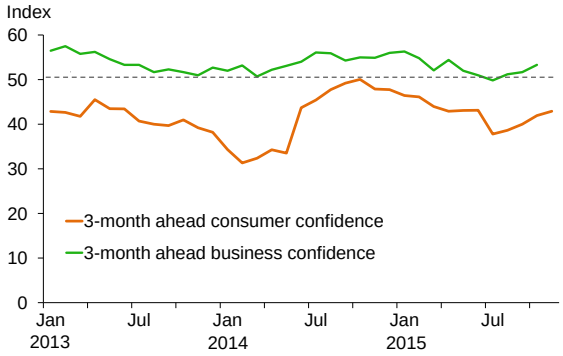
Chart 2.6 Public spending



Source: Bureau of Budget and Fiscal Policy Office

The government economic stimulus package, introduced in steps since September 2015, partly helped both consumer and business confidence **(Chart 2.7)**. The stimulus included measures to boost the grass-root economy at the village and sub-district levels, lending and tax measures to support SMEs, and tax benefits to encourage new domestic investment. Moreover, the government introduced measures to support activity in the **real estate sector**, providing housing loans, tax rebates, and reduction of transfer and mortgage fees. These short-term measures should help reduce excess supply of residential units by incentivizing homebuyers to sooner purchase and transfer properties, especially condominiums priced below 3 million baht which account for about 60 percent of the total unsold units outstanding in Bangkok and its vicinities.

Chart 2.7 Consumer Confidence Index



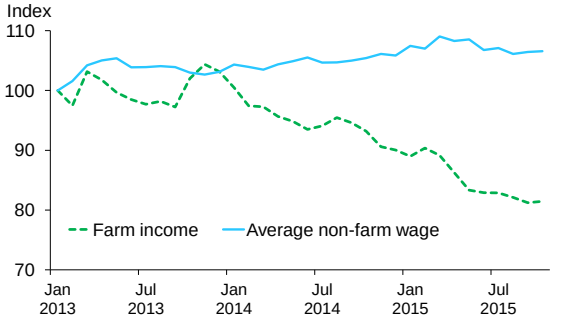
Source: Ministry of Commerce, calculations by Bank of Thailand

Private consumption recovered gradually in line with increased spending on necessities and services, but spending on durables remained subdued due to weak income and elevated household debt.

Private consumption in the non-durables category, especially necessities and services such as telecommunications and transportation, picked up slowly. While household purchasing power was bolstered by higher consumer confidence in the economic outlook, due to the government's economic stimulus as well as persistently low inflation and oil prices, spending on durables remained generally weak. That was because (1) farm income contracted due to depressed farm prices, especially rubber prices, and also due to a fall in rice production resulting from the drought; non-farm income remained flat **(Chart 2.8)**; (2) financial institutions remained cautious about lending; and (3) household debt remained elevated.

Chart 2.8 Real Income

(seasonally adjusted, 3-month moving average, Jan 2013 = 100)



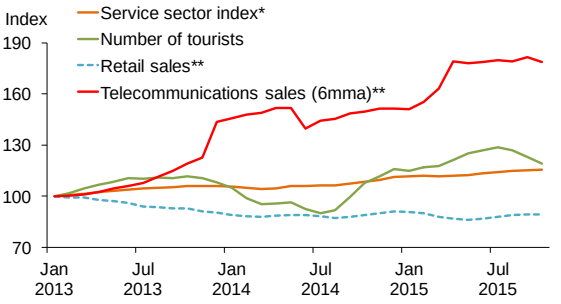
Source: National Statistical Office and Office of Agricultural Economics, calculations by the Bank of Thailand.

While spending on automobiles edged up in Q3, that was partly attributable to the launching of new commercial car models and the accelerated purchase of vehicles before the increase in excise tax on automobiles takes effect in 2016.

Household spending on services were well maintained and helped underpin the service sector. This was reflected in the continuing rise in revenues of telecommunications firms (**Chart 2.9)**, consistent with increased data usage resulting from changes in consumer lifestyles. In addition, the government auctioned the 4G spectrum licenses in late 2015. Winning firms are expected to start offering the services in early 2016.

Chart 2.9 Service sector index

(seasonally adjusted, 3-month moving average, Jan 2013 = 100)



Note: *comprises VAT levied on hotels and restaurants, and sales of transportation services.
**indicators not included in private consumption index.

Source: The Revenue Department, Department of Tourism, Department of Lands, calculations by Bank of Thailand

Exports remained weak in both quantity and price terms due both to cyclical factors associated with the slowdown in major trading partners' economies and to existing structural problems.

Cyclical factors in the global economy had more pronounced impacts on Thai exports. Although the G3 economies, such as the U.S., continued to show signs of recovery, major trading partners, especially ASEAN, appeared to be harder hit by the slowdown in China. Moreover, existing structural problems weighed on Thailand's manufacturing and exports. One example of merchandise exports that suffer from a loss of competitiveness is the hard disk drives (HDDs). Global demand for HDDs has slowed down because consumers switch to new products, such as tablets and smartphones that require solid state drives (SSDs), which Thailand is capable of manufacturing only in small quantities. Apparel exports have also suffered from the European Union's withdrawal of GSP privileges in early 2015. These structural constraints led to the continuing decline in exports values of HDDs and

apparels **(Chart 2.10)**. As a result, Thailand's overall export volume has continued to contract.

Moreover, plunging oil prices, combined with lower commodity prices following weakening Chinese growth, have led to a sustained decline in Thai export prices. These falling prices were particularly acute for commodity exports and oil-related products—such as agricultural products, petroleum, petrochemicals and chemicals—which altogether account for 14.5 percent of total export value **(Chart 2.11)**.

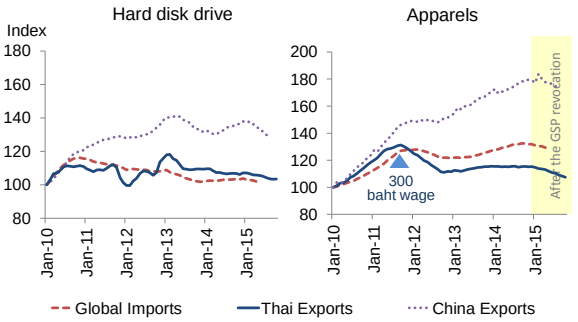
Nevertheless, exports of some product categories still expanded. For example, electronic parts and components used in China's manufacturing of new smart phone and tablet models registered a solid and sustained expansion. Thailand's exports of automobiles also recorded a strong growth, consistent with increasing hours worked **(Chart 2.12)**, as a result of a change in production line for new car models and also by virtue of having multiple export destinations that helped cushion the impacts of the global economic slowdown.

Private investment remained sluggish given ample excess production capacity.

Although private consumption gradually picked up, weak exports led to ample excess production capacity especially for export-oriented manufacturing, and that eliminated the needs for new investment. Consequently, private investment contracted, with the value of imported machinery remaining low in several industries. Factors that weighed down private investment included: (1) a slowdown in external demand especially from China and ASEAN, (2) cautious lending by financial institutions especially to SMEs, and (3) the government's megaprojects that were still in

Chart 2.10 Exports of merchandise indices

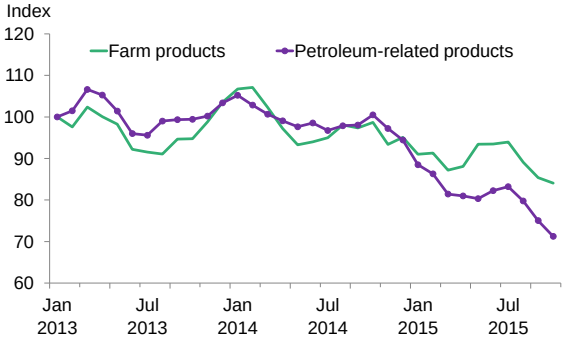
(seasonally adjusted, 12-month moving average, Jan 2013 = 100)



Source: Customs Department and Trademap, calculations by Bank of Thailand.

Chart 2.11 Commodity exports value index

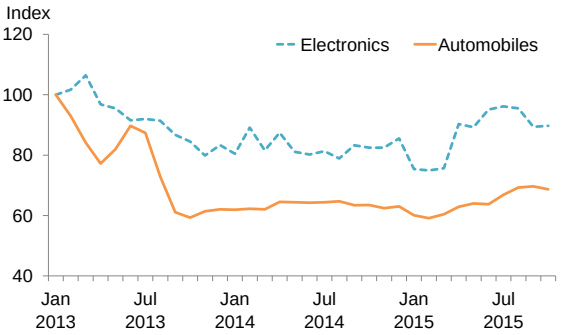
(seasonally adjusted, 12-month moving average, Jan 2013 = 100)



Note: *comprises petroleum, petrochemicals and chemicals.

Chart 2.12 Working hours index

(seasonally adjusted, 3-month moving average, Jan 2013 = 100)



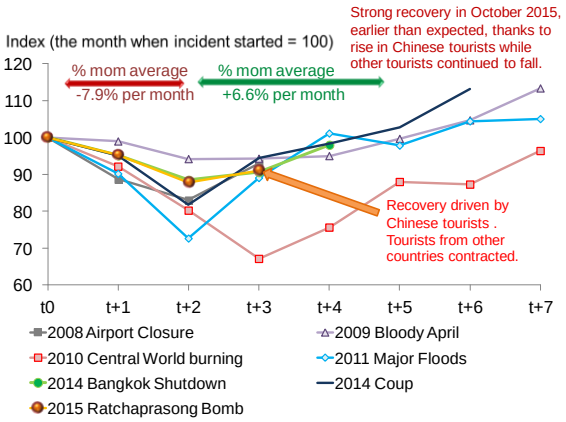
Source: Labour Force Survey, National Statistical Office.

the initial phases. New investment was thus restricted only to some sectors, for example, the telecommunications industry following the 4G spectrum auction.

Tourism saw a slower growth in the near term but started to show positive signs.

Tourism continued to lend support for the economy despite some temporary drops in the aftermath of the bombing incident in Bangkok. There were stronger signs of recovery, particularly in the number of Chinese visitors. Tourist arrivals rebounded in October 2015 **(Chart 2.13)**, having earlier declined for two consecutive months. Going forward, tourist arrivals were expected to gradually rebound, largely thanks to the Chinese tourists, during the high season. Nonetheless, downside risks to tourism need to be monitored, stemming from intensifying geopolitical risks and the potential impacts of the U.S. downgrade of Thailand's aviation safety standards.

Chart 2.13 Index of foreign tourists
Following past violent incidents



Source: Department of Tourism, calculations by Bank of Thailand.

2.3 Production costs and price conditions

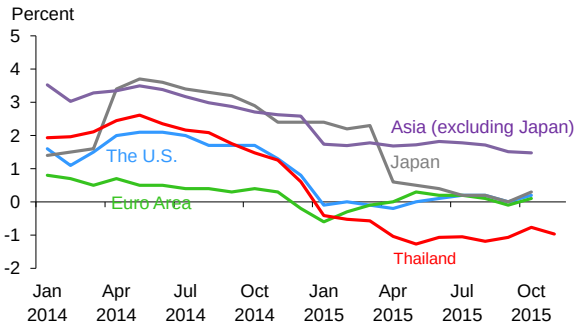
Headline inflation for the first 11 months of 2015 averaged a negative 0.90 percent, owing mainly to supply-sided factors associated with the substantial fall in global crude prices relative to the same period last year. Meanwhile, demand-pull inflationary pressures edged up consistent with the gradual recovery of domestic demand. Looking ahead, the MPC expects headline inflation to rise gradually and turn positive during the first half of 2016, as the base effects of high oil price dissipate and domestic demand rebounds.

Recent structural changes in global oil production, especially the development of new drilling technologies in the U.S., increased the global oil supply. The excess supply was reflected in growing crude oil reserves of non-OPEC countries, while OPEC members continued to maintain high levels of production. At the same time, global demand for oil remained weak in line with weaker growth in China and Asia and a modest recovery in G3. Consequently, global crude oil prices and headline inflation readings in a number of countries were subdued **(Chart 2.14)**.

Thailand's headline inflation in the first 11 months of 2015 averaged -0.90 percent, primarily due to lower domestic energy prices in tandem with plunging global crude oil prices **(Chart 2.15)**. For the first 11 months of the year, domestic energy prices on average fell by 14.6 percent compared with the same period last year. However, for the first two months of Q4, domestic energy prices contracted at a slower rate, as the base effects of high oil prices gradually waned. Cooking gas prices edged up in line with the government's elimination of gas price controls. Raw food prices picked up slightly in line with prices of fruits and vegetables following volatile weather.

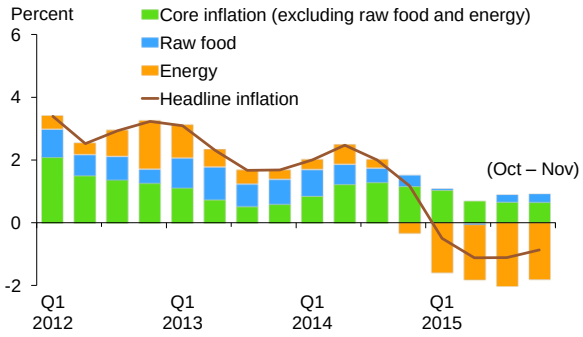
Nevertheless, negative headline inflation continued to persist. Although the baht depreciation since 2015 Q2 pushed up headline inflation somewhat, it could not fully offset the effects of lower crude oil price. Moreover, exchange rate pass-through to inflation remained small. A 1 percent baht depreciation is estimated to cause headline inflation to rise by only 0.06 percent, with most of the effects occurring through retail oil prices (which move in line with imported crude prices and the exchange rate). In contrast, the

Chart 2.14 Domestic and foreign headline inflation



Note: Foreign headline inflation calculated from unweighted averages
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, CEIC, and calculations by Bank of Thailand

Chart 2.15 Contributions to headline inflation



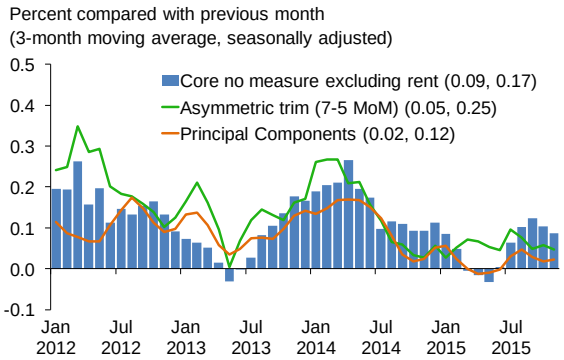
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

impacts of the baht depreciation on other prices are marginal because goods that rely on imported raw materials account for only a small portion (about 16 percent) of the CPI basket.[3/] In addition, not only pricing of necessity goods by firms is largely regulated by the Ministry of Commerce, some firms have hedged against exchange rate risks according to surveys of business operators. Together with the weak recovery of domestic demand and a high degree of domestic competition, firms cannot fully pass on higher costs to consumers.

Despite declining cost pressures resulting from lower crude oil prices, demand-pull pressures picked up slightly from increasing household consumption in some categories. However, demand-pull pressures remained low overall due to concerns over future income, especially for households hit by drought and those working in the manufacturing and export sectors, both of which faced greater risks from a slowing global economy. Household debt also remained elevated. Thus, underlying inflation indicators remained low **(Chart 2.16),** with core inflation for the first 11 months of 2015 averaging 1.08 percent. That is below the long-term historical average.

Nonetheless, deflation risks remained limited due to sustained demand growth and core inflation remaining stable and positive. Short-term inflation expectations were stable due to subdued global crude oil prices. Medium-term inflation expectations remained close to the medium-term inflation target of 2.5 percent: while 1-year-ahead

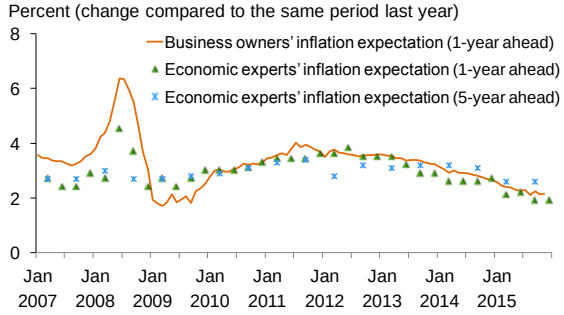
Chart 2.16 Underlying Inflation



Note: Inside parentheses, figures on the left indicate Nov 2015 %MoM_SA (3mma), while those on the right show long-term average (2005-2014). (Read more in FAQ Issue 93, Underlying Inflation)

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Chart 2.17 Public inflation expectations



Note: Business owners' inflation expectation surveyed by Bank of Thailand
Economic experts' inflation expectation surveyed by Consensus Economics

Source: Business Sentiment Index Survey by Bank of Thailand and Asia Pacific Consensus Forecast

[3/] The extent of imported raw materials in the CPI basket is calculated using information from the input-output (I-O) table and is assigned weights according to the CPI basket.

inflation expectations of professional forecasters, surveyed by Consensus Economics, were 1.9 percent, the 5-year-ahead expectations were 2.6 percent **(Chart 2.17)**.

Going forward, the MPC judged that inflationary pressures should continue to be subdued and headline inflation should pick up slowly and turn positive in the first half of 2016 due to the waning base effect of high oil prices and gradual economic recovery.

Table 2.1 Quarterly inflation

Unit: percent	2013	2014		2015			
		Q3	Q4	Q1	Q2	Q3	Oct-Nov
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index	2.18	2.00	1.11	-0.50	-1.12	-1.11	-0.87
• Core Consumer Price Index	1.00	1.79	1.65	1.47	0.97	0.93	0.92
• Raw food	5.54	2.78	1.88	0.30	-0.40	1.48	1.66
• Energy	4.79	2.20	-3.11	-12.95	-14.16	-16.42	-15.20
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index	-	0.0	-0.2	-0.8	-0.1	0.0	-
• Core Consumer Price Index	-	0.3	0.3	0.3	0.1	0.3	-
• Raw food	-	-0.5	0.5	0.0	-0.3	1.4	-
• Energy	-	-1.1	-3.8	-9.0	-0.8	-4.2	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce. Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

Assessing corporate debt risks in emerging markets and Thailand

The conduct of monetary policy by major central banks following the global financial crisis of 2008 has led to greater liquidity in the global financial markets. Some liquidity subsequently flew to emerging market economies, causing the borrowing costs and risk premiums in these markets to significantly decline from the pre-crisis levels. This was one of the factors that caused firms in most emerging markets, especially in East Asia and Latin America, to increase their leverage, as shown by the rise in the corporate debt-to-GDP ratio **(Chart 1)**.

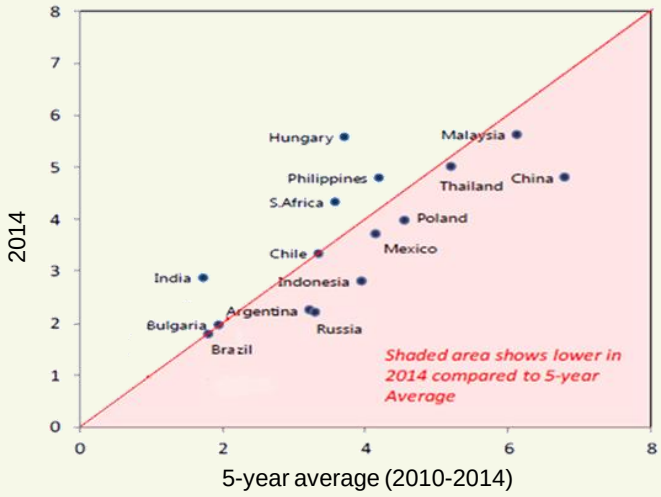
Chart 1 Changes in corporate debt from the global financial crisis to present



Source: GFSR, April 2015

Now, with the monetary policy normalization by the Fed, some capital flows have been redirected out of emerging markets. This has driven up costs of capital faced by firms, as reflected in the higher yields on long-term government bonds in some countries, and caused emerging market currencies to weaken against the U.S. dollar. As a result, external debt liability, measured in local currencies, have increased. Moreover, subdued global commodity prices and a lower growth outlook, especially in commodity-producing and commodity-exporting countries, have weakened the corporate sector's ability to service debt. **Chart 2** shows a fall in the interest coverage ratio (ICR) below its historical average in most emerging markets—and to a particularly low level in some countries. In view of the rapid expansion of corporate leverage in recent years in some emerging markets, the current situation raises concerns over financial stability in those countries.

Chart 2 Interest coverage ratio (ICR), 2014

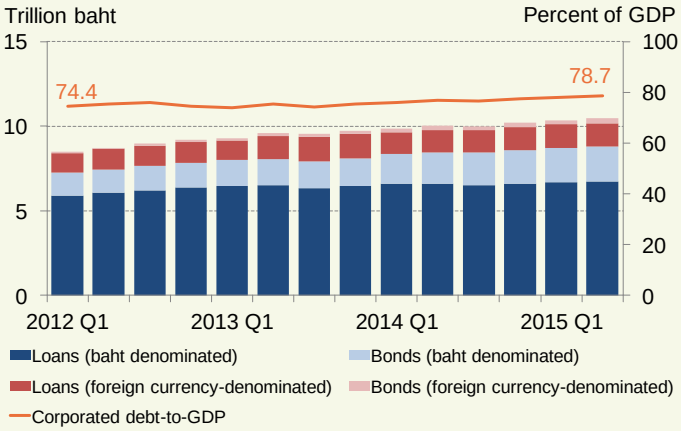


Source: IMF working paper: "Stress Testing Corporate Balance Sheets in Emerging Economies" (Sep 2015)

As for Thailand, the extent of risks posed by external corporate debt remain limited due to the following reasons:

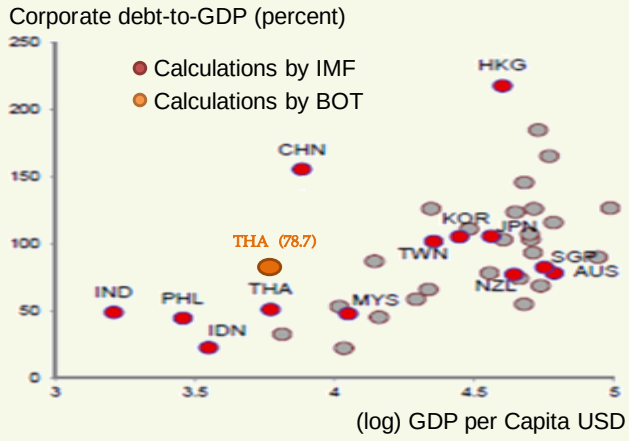
• Overall corporate debt has not expanded rapidly and the rate of increase is quite moderate compared with other countries, especially when taking into consideration the level of financial sector development. Thailand's corporate debt-to-GDP for non-financial sector was 78.7 percent in 2015 Q2, a modest increase from 74.7 percent at the end of 2014 **(Chart 3)**. This is consistent with the finding in the IMF's *Global Financial Stability Report* (GFSR), which depicts a 7 percent growth in Thailand's corporate debt from during 2007-2014[1/] which is moderate relative to those in other emerging markets. In addition, if GDP per capita is used as a proxy for the level of financial sector development, Thailand's corporate debt-to-GDP ratio is not high compared with countries with similar levels of GDP per capita[2/] **(Chart 4)**.

Chart 3 Thailand's corporate debt-to-GDP



Source: Bank of Thailand

Chart 4 Corporate debt-to-GDP, Thailand and other countries



Note: BOT data covers all loans and bonds issued by the corporate sector. IMF data is based on BIS data, which includes loans and bonds held by other depository corporations (ODC).

Source: IMF, Asia and Pacific Department, Presentation, September 2015

• The impacts of the baht depreciation on external corporate debt burden remain limited because Thailand's foreign currency-denominated corporate debt is not substantial relative to that of other emerging markets.[3/] Moreover, a vast majority of Thailand's external corporate debt is incurred by large corporations that can effectively manage exchange rate risks. Some of

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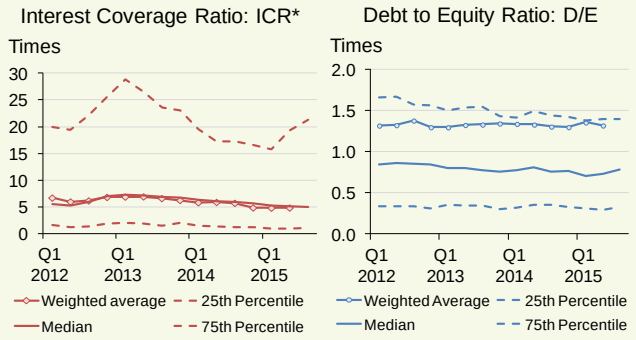
[1/] Data from GFSR, April 2015.

[2/] Data on Thailand's corporate debt, which include loans and corporate bonds issued by non-financial firms and held by domestic and foreign entities, are compiled by the BOT. Meanwhile, Data on other countries' corporate debt are obtained from the IMF, which in turn uses the database of the Bank for International Settlements (BIS). This database includes loans and bonds held only by depository institutions, the corporate debt figures of other countries may be below the actual levels.

[3/] Data from GFSR, October 2015.

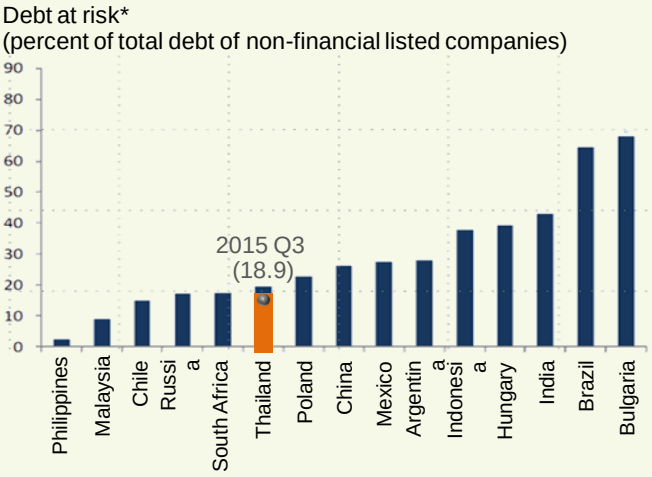
these businesses receive exports earnings and investment returns in foreign currencies—a natural hedge for their increased external debt burden. Meanwhile, most large businesses with foreign currency exposures continue to engage in financial hedges (i.e., financial transactions aimed at protecting against exchange rate risks) and tend to increase the extent of hedging if faced with the weakening trend of the baht.[4/]

Chart 5 Financial indicators of non-financial businesses listed in the stock exchange (SET and MAI)



Note: *Annualized calculation.
Source: Stock Exchange of Thailand, calculations by the Bank of Thailand.

Chart 6 Debt at risk, Thailand and other countries, 2014



Note: *Debt at risk is the ratio of debt held by companies with ICR lower than 1.5 to the total debt held by non-financial companies listed in SET and MAI, based on financial statements of listed companies in 2015 Q3.
Source: Julian T.S. Chow (2558) "IMF working paper: "Stress Testing Corporate Balance Sheets in Emerging Economies" (Sep 2015)", Stock Exchange of Thailand, calculations by the Bank of Thailand.

• Corporate financial positions remain sound. Although the earnings of non-financial companies listed on the stock exchange (SET and MAI) declined somewhat amid sluggish economic recovery, firms' ability to service debt remains intact. The current ICR indicates that profits are able to cover five times the interest expenses, while debt-to-equity ratio (D/E ratio) remains at low levels **(Chart 5)**. Moreover, the extent of debt considered at risk (leading to ICR falling below 1.5, which signals higher risks in meeting interest payment obligations in the future) in the Thai corporate sector has remained quite steady since the global financial crisis. In addition, most of the debt in this category largely belong to the same firms. As of 2015 Q3, Thailand's debt-at-risk stood at 18.9 percent of the total corporate debt of firms listed at the stock exchange, a modest level relative to debt-at-risk ratios for other emerging markets **(Chart 6)**.

In conclusion, since the global financial crisis, firms in emerging markets have significantly increased their leverage, and this prompts concerns over financial stability of the corporate sector in these countries against the backdrop of the slow global recovery and the prospect of higher funding costs in the global financial market following the Fed's monetary policy normalization. Nevertheless Thailand's corporate debt has not increased

[4/] For more detailed analysis, see Box on "Baht depreciation: implications on external debt and external stability" in the Monetary Policy Report, September 2015.

significantly. Risks arising from external debt remain limited and do not appear to undermine overall financial stability given sound financial positions of and effective exchange rate risk management practices by listed companies. Nevertheless, looking ahead, uncertainty over economic recovery and potentially greater financial volatility potentially affect debt-servicing ability of some businesses, particularly small firms or those with weak financial positions. Therefore, the MPC will closely monitor these developments and the adjustments of the Thai business sector in response to these challenges.

Monetary and Financial Stability

3. Monetary and Financial Stability

Overall monetary conditions continued to be accomodative. Costs of funds financed by commercial banks and capital markets remained broadly unchanged from the previous quarter. However, money market rates and exchange rates were somewhat volatile, driven by shifts in investor expectations over the timing of the federal funds rate increase.

A moderation in income growth caused by the slow and uneven economic recovery of the Thai economy adversely affected financial health of households and firms, especially small businesses. Nonetheless, commercial banks remained sound, with high levels of loan loss provision and capital buffers acting as a cushion against any risks from deteriorating credit quality. The property sector slowed down, but the government's real estate stimulus measures should generate demand and reduce some unsold inventories. The search-for-yield behavior among investors in a prolonged low interest rate environment remained visible, as partly shown by the extent of investment by retail investors in the stock market that edged higher and by the P/E ratio that remained above the historical average. Fiscal stability remained sound, with a sufficient level of treasury cash balance to cover below-target revenue collections as well as with a manageable level of public debt. Nevertheless, going forward, public debt would likely rise due to the new borrowings to cover rising budget deficit, investments in transportation infrastructures, and increasing fiscal liabilities associated with economic stimulus packages.

3.1 Financial markets

Global financial markets

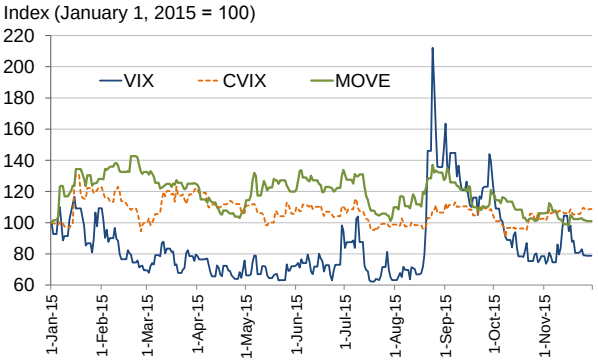
After a rise in early September, volatility in the global financial markets declined as shown by the VIX index (Volatility Index)[1/] for the U.S. equity market and the MOVE (Merrill Lynch Option

[1/] VIX is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P 500.

Volatility Estimate)[2/] for the U.S. treasury market. Both indices fell from the levels observed in early September. However, volatility in the foreign exchange markets remained elevated as shown by the CVIX (Currency Volatility Index)[3/] (**Chart 3.1)**. These volatilities in the financial markets were caused by the uncertainty over the timing of federal funds rate increase at the same time as the European Central Bank and the Bank of Japan looked set to continue with monetary policy easing. Furthermore, China's slowdown added to market uncertainty through the negative spillovers to economies closely linked to China, especially emerging markets, by way of falling exports and declining commodity prices.

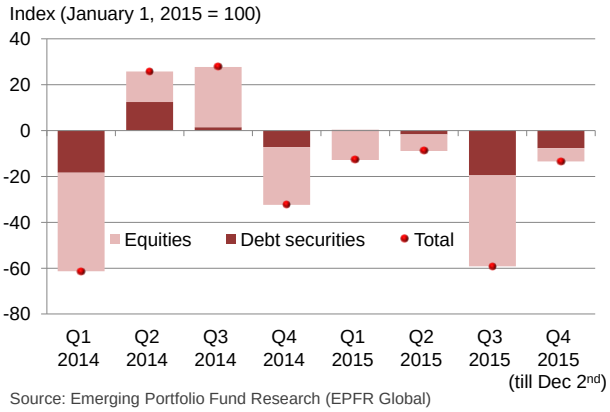
Such global factors, combined with the slowdown in emerging market economies, triggered capital outflows from equity and bond markets in the third quarter of 2015 **(Chart 3.2).** Going forward, differing macroeconomic fundamentals and monetary policy divergence between countries would give rise to further volatilities in financial markets and movements in capital flows and asset prices.

Chart 3.1 Volatility indices



Source: Bloomberg

Chart 3.2 Capital inflows to emerging markets



Source: Emerging Portfolio Fund Research (EPFR Global)

Thailand's financial markets

Money market rates stabilized in line with the policy interest rate. Funding costs of business in the capital market also remained steady, although they fluctuated from time to time along with changes in investor expectations over the timing of federal funds rate increase.

[2/] MOVE is an indicator for bond market volatility, measured by implied volatility of U.S. treasury bonds.

[3/] CVIX is an indicator for currency volatility, measured by implied volatility of 9 major currencies.

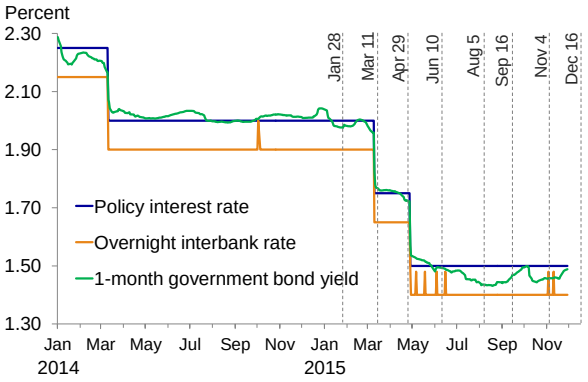
Money and bond markets

From October to December 2015, money market rates stabilized in line with the policy interest rate **(Chart 3.3).** Similarly, overall funding costs of business in the bond market remained steady from the previous quarter, although they fluctuated along with the medium- and long-term Thai government bond yields, which in turn were influenced by changes in investor expectations over the timing of federal funds rate increase **(Chart 3.4)**. Meanwhile, credit spreads were largely stable **(Chart 3.5).**

In October 2015, medium- and long-term Thai government bond yields fluctuated and declined in tandem with the U.S. treasury bond yields. The decline was a result of investors' expectations at the time that the Federal Reserve would postpone its first policy rate increase, after U.S. economic data, such as employment figures, came in weaker than expected. The FOMC statement in September 2015 cited concerns about the economic recovery if the policy rate were lifted too soon. As a result, investors returned to the Asian bond markets, including Thailand.

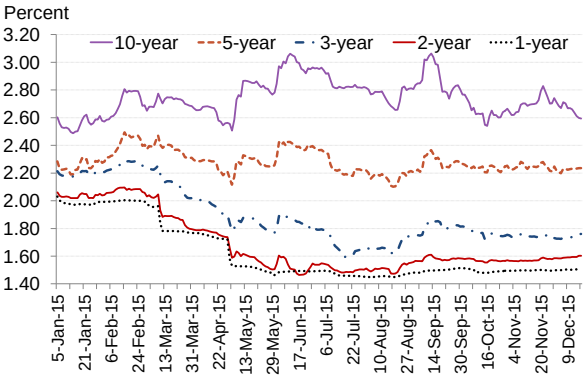
However, from November to December, better-than-expected data releases and communication by the FOMC about the monetary policy outlook led to a greater clarity that the Federal Reserve would likely raise its policy rate at the December meeting. Investors, therefore, reduced their holdings of emerging market bonds, leading to rising bond yields in emerging markets, including Thailand. Furthermore, after the ECB announced monetary policy easing that fell short of market expectations, together with crude oil prices that dropped further sharply, Thai government bond yields became

Chart 3.3 Money market interest rates



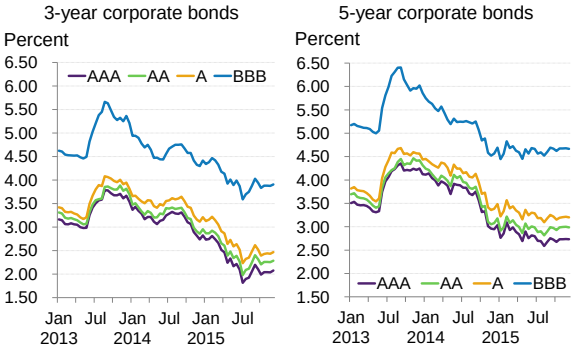
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.4 Government bond yields in 1 to 10-year sectors



Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.5 Thai corporate bond yields



Source: Thai Bond Market Association (Thai BMA)

more volatile, moving in line with U.S. treasury yields **(Chart 3.6 and 3.7).**

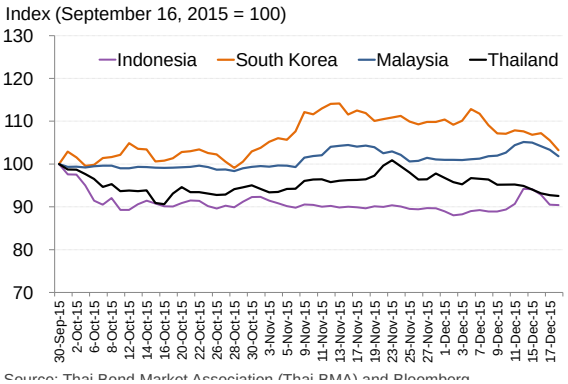
Going forward, cross-border capital flows and financial conditions would likely become more volatile, as central banks pursue increasingly divergent monetary policies at the same time as the global economic recovery remained fragile. The Committee would continue to closely monitor developments in financial markets and their implications for liquidity and costs of funds.

Investment in financial assets

Search-for-yield behavior continued to increase in the prolonged low interest rate environment as indicated by rising investment in risky assets—such as domestic equities, non-rated bonds, and foreign securities—relative to deposits. Nevertheless, the attendant risks associated with search-for-yield behavior to overall financial stability were limited.

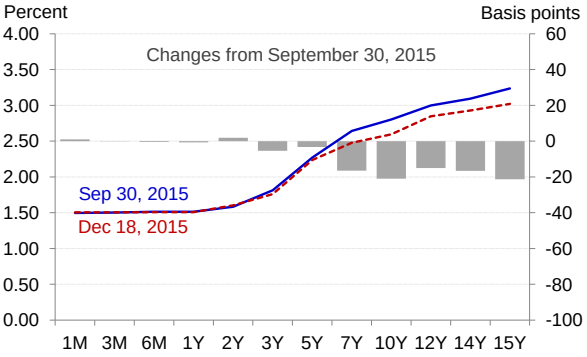
Equity market trading slowed as reflected by the decline in the Stock Exchange of Thailand (SET) Index over the past three months. The index stood at 1,284.92 on December 18, 2015, down from 1,349.0 at the end of 2015 Q3 **(Chart 3.8)** following the year-over-year drop in earnings of listed companies due to the gradual economic recovery. Non-resident investors continued to be net sellers on the back of the region's economic slowdown and expectations about monetary policy normalization by the Federal Reserve. On the other hand, resident investors, both institutional and retail, were net buyers. For the MAI, after the Securities and Exchange Commission (SEC) implemented measures to curb irregular trading activities at the start of the year, investment by retail investors continued to decline, suggesting that speculative activities subsided **(Chart 3.9)**.

Chart 3.6 10-year government bond yields



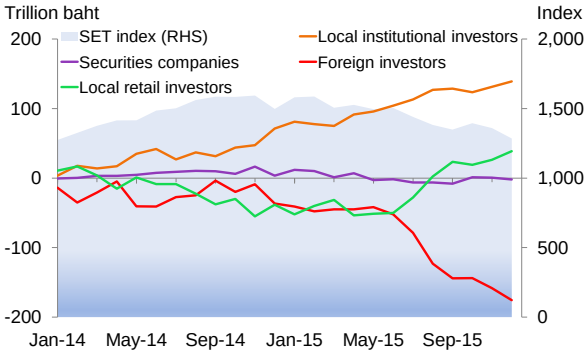
Source: Thai Bond Market Association (Thai BMA) and Bloomberg

Chart 3.7 Changes in government bond yield curves



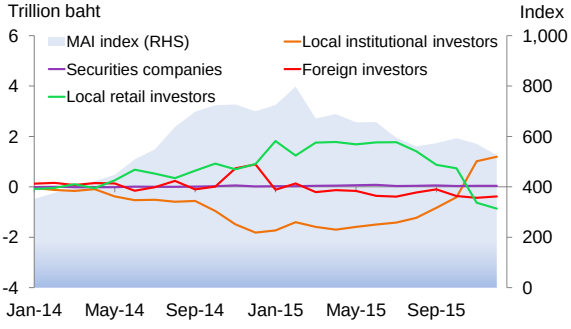
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.8 The Stock Exchange of Thailand Index and net buy classified by investor types



Source: Stock Exchange of Thailand (latest data on December 18, 2015)

Chart 3.9 The Market for Alternative Investment (MAI) and net buy classified by investor types



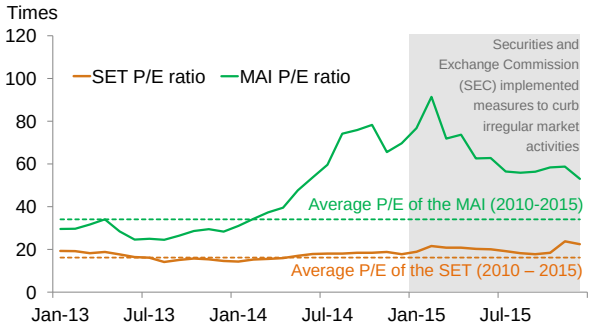
Source: Stock Exchange of Thailand (latest data on December 18, 2015)

Although stock indices fell, prices remained high relative to the earnings per share, with the price-earnings (P/E) ratio of the SET and the MAI being above the historical averages **(Chart 3.10).** Going forward, there would be a risk that prices may continue to fall further, especially if a delayed economic recovery were to affect the performances of listed companies. Nevertheless, risks to financial stability should be limited, given a small proportion of trading was funded by margin loans, which accounted for only 12 percent of total trading as of November 2015.

With regard to the domestic bond market, an increasing number of companies recently issued non-rated bonds **(Chart 3.11)**, consistent with searching for yield in excess of deposit rates. However, risks from non-rated bonds to financial stability were limited, because they accounted for only a small proportion of overall private sector bonds outstanding and were offered only to large accredited investors.

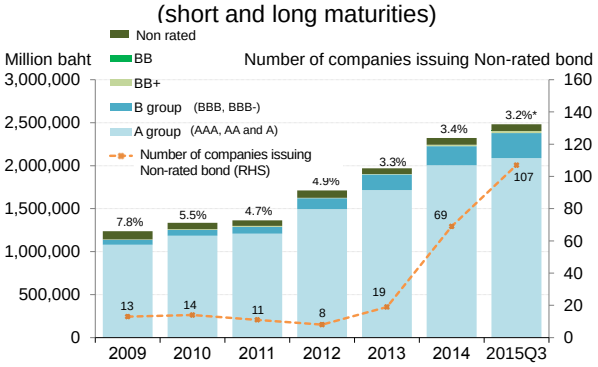
The rise in the search for yield behavior in foreign securities by institutional investors and mutual funds was also observed **(Chart 3.12)**. Some of the funds were invested in countries with high credit risks, such as Brazil, whose credit rating was lowered to non-investment grade in September 2015. However, risks to overall financial stability were limited, since most of the funds were invested in countries with investment-grade ratings. In addition, the risks from investment in the Brazilian assets were small, because most were invested in term funds of short maturity and mainly held by large accredited investors (AI).

Chart 3.10 Price - Earnings Ratio (P/E)* of SET and MAI



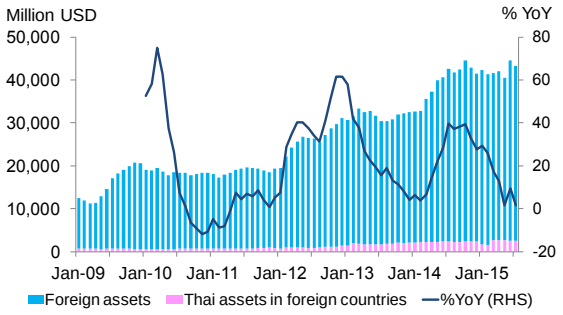
Note: *Earnings Per Shares: EPS over the past 12-month period
Source: Stock Exchange of Thailand (latest data on December 18, 2015)

Chart 3.11 Corporate bonds outstanding (short and long maturities)



Note : *Percent of Non-rated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Chart 3.12 Outstanding investment in foreign assets*



Note: *Foreign Investment Funds (FIF), insurance companies, and government pension fund account for over 90 percent of outstanding investment in foreign asset.
Source: Bank of Thailand

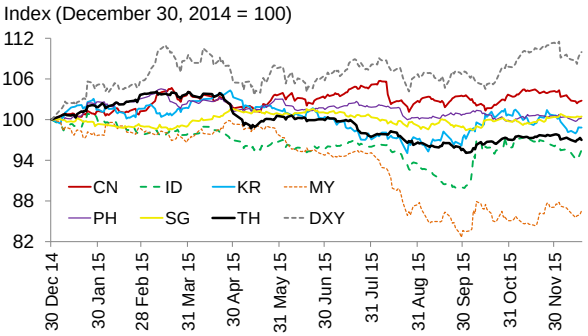
Foreign exchange market

The baht appreciated against major currencies between late-September and mid-October 2015. During that period, investors pushed back the timing for the federal funds rate increase and returned to equity and bond markets in Asia, including Thailand, following disappointing U.S. economic data, especially the employment figures. The baht appreciation was in line with most currencies in the region **(Chart 3.13).**

From mid-October to mid-December 2015, the baht significantly strengthened against the euro, due in part to additional monetary policy easing by the ECB to cope with subdued core inflation and to support economic recovery. As a result, the Nominal Effective Exchange Rate (NEER) appreciated by 1.2 percent on December 18, 2015 relative to the end of 2015 Q3. The Real Effective Exchange Rate (REER) rose along with the NEER **(Chart 3.14).**

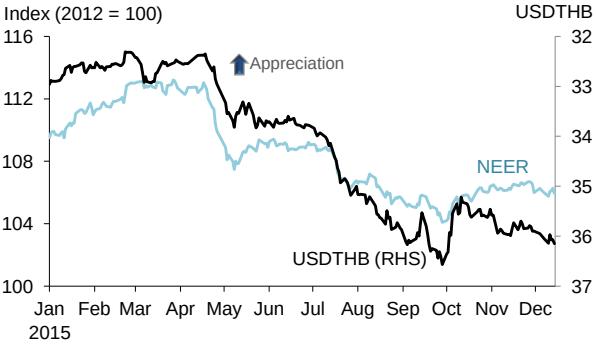
In contrast to the NEER appreciation, the baht against the U.S. dollar weakened since mid-October after U.S. core inflation and employment data came out much better than expected, prompting investors to widely anticipate the first U.S. rate hike to be in December. Therefore, there were minimal movements in currencies in the region when the federal funds rate increase was finally announced on December 16. As of December 18, the Thai baht closed at 36.15 baht per U.S. dollar, a depreciation of 2.7 percent from October 15[4/] (nonetheless appreciating 0.4 percent from the end of the third quarter).

Chart 3.13 Regional exchange rate indices



Note: Based on average trade weights between 2008 and 2010 from BIS
Source: Reuters, calculations by the Bank of Thailand
(latest data on December 18, 2015)

Chart 3.14 Movements in USDTHB and exchange rate index

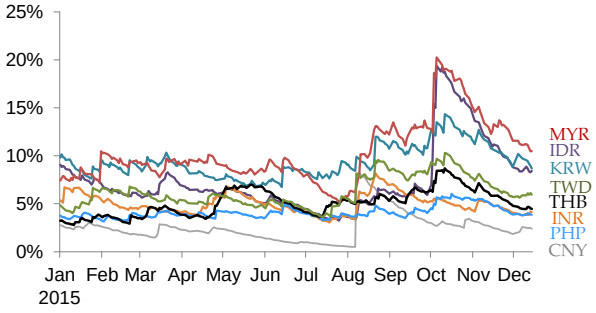


Source: The Bank of Thailand (latest data on December 18, 2015)

[4/] On October 15, 2015 the U.S. released higher-than-expected core inflation, and an ECB executive board member commented that core inflation in the euro area was substantially below target, calling governments to play a more active role in stimulating the economy.

The volatility of the baht from October to December 18, 2015 averaged 6.4 percent, up from the average of 5.2 percent in the previous quarter. Nevertheless, compared with most currencies in the region, the volatility of the baht remained moderate **(Chart 3.15).**

Chart 3.15 Volatility of the baht and regional currencies



Note: Volatility is computed by Exponentially Weighted Moving Average (EWMA) method

Source: The Bank of Thailand and Reuters

3.2 Financial institutions

Private sector costs of funds and returns on deposit remained unchanged in line with the policy rate. Total funds raised by the private sector slowed from 2015 Q2. A gradual economic recovery led to subdued new deposits through special deposit products, resulting in weak deposit growth.

Interest rates, credits, and deposits

From 2015 Q3 to November, funding costs for commercial banks stabilized given that there was no change in the policy rate **(Table 3.1).**

Table 3.1 Benchmark loans and deposits rates of commercial banks*

Percent	2013	2014			2015			
		H1	Q3	Q4	Q1	Q2	Q3	Dec 18
12-month deposits								
Average of the 4 largest commercial banks**	2.23	1.73	1.73	1.73	1.53	1.50	1.46	1.41
Average of remaining banks ***	2.61	2.22	2.19	2.22	1.93	1.75	1.72	1.59
Minimum lending rate (MLR)								
Average of the 4 largest commercial banks	6.84	6.75	6.75	6.75	6.63	6.51	6.51	6.51
Average of remaining banks	7.62	7.51	7.45	7.44	7.37	7.32	7.31	7.25
Minimum retail rate (MRR)								
Average of the 4 largest commercial banks	7.96	8.05	8.08	8.08	8.08	7.86	7.87	7.87
Average of remaining banks	8.77	8.70	8.63	8.63	8.62	8.54	8.54	8.54

Note: *Benchmark rates averaged across commercial banks at the end of the periods.

** Four largest commercial banks namely BBL, KTB, KBANK and SCB.

*** Starting in August 2014, remaining banks include Bank of China (Thailand). From June 2015, they include ANZ (Thailand). And since October 2015, they also include Sumitomo Mitsui Trust Bank (Thailand).

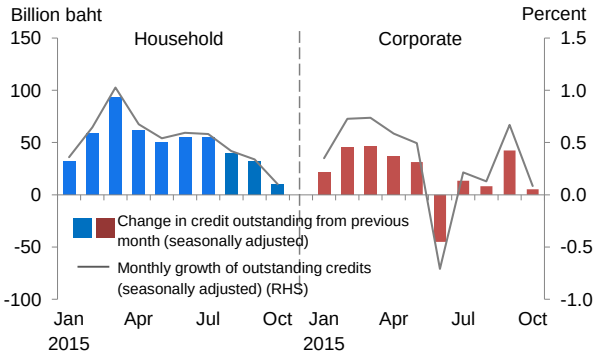
- On December 18, 2015, the average 12-month deposit rates fell substantially from the end of 2015 Q3, partly due to the inclusion of Sumitomo Mitsui Trust Bank (Thailand), whose rates were relatively low.

New household credits during August–October fell relative to 2015 Q2 **(Chart 3.16),** as households remained cautious in their spending. Moreover, the credit conditions survey for 2015 Q3 indicated that financial institutions were cautious about lending because of concerns over households' repayment ability given the slow economic recovery and elevated household debt. Housing loan growth also slowed, owing in part to a temporary factor as home buyers awaited a greater clarity on the government's real estate stimulus package due to be announced in October.

The corporate sector continued to raise funds between August and October, although the amount was slightly smaller than that raised during 2015 Q2 **(Chart 3.17).** Seasonally adjusted new credits to corporate sectors grew somewhat, but it did not amount to a significant recovery. New loans were concentrated in the real estate and renewable energy sectors, while manufacturing and trading sectors saw a decline in loan issuance **(Chart 3.18).** The drop was particularly large among SMEs due to weak demand for credits and tightened credit standards. The credit conditions survey also indicated financial institutions would tighten credit standards for SMEs further in 2015 Q4. In addition, large firms continued to finance through bond issuance instead of commercial bank loans. Between August and October, new bonds were mainly issued by real estate developers and telecommunication service providers.

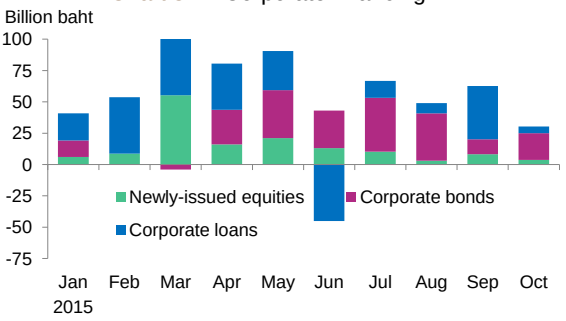
Outstanding deposits at the end of October increased slightly from the end of July thanks to a rise in household deposits at commercial banks and specialized financial institutions (SFIs). Nevertheless, growth in new deposits (including bills of exchange, B/E) remained low **(Chart 3.19).** Given stagnant demand for credits, commercial

Chart 3.16 New private credits



Source: Bank of Thailand

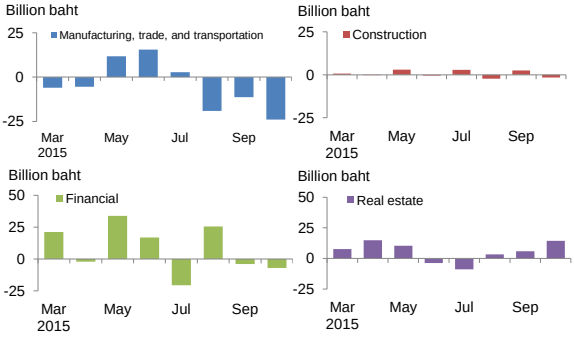
Chart 3.17 Corporate financing*



Note: *Monthly change in corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.

Source: Bank of Thailand and Thai BMA

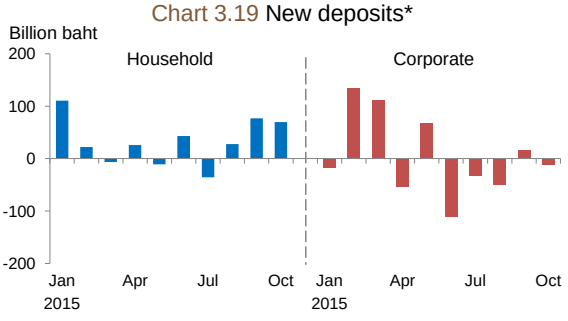
Chart 3.18 Newly-issued corporate loans*



Note: *Monthly change in commercial bank loans for corporates

Source: Bank of Thailand

Chart 3.19 New deposits*



Note: *Change in outstanding deposits at depository institutions (excluding Bank of Thailand). The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.

banks saw little need to raise new deposits, as reflected by continued reduction in interest rates on special deposit products **(Chart 3.20).** Meanwhile, some savers moved deposits toward higher-yielding instruments such as insurance products and mutual funds that invested in domestic as well as foreign assets. Due to relatively stable credits and deposits, the loan to deposit ratio (including B/E) stayed roughly unchanged between August and September, before dipping in October owing importantly to credit slowdown **(Chart 3.21).**

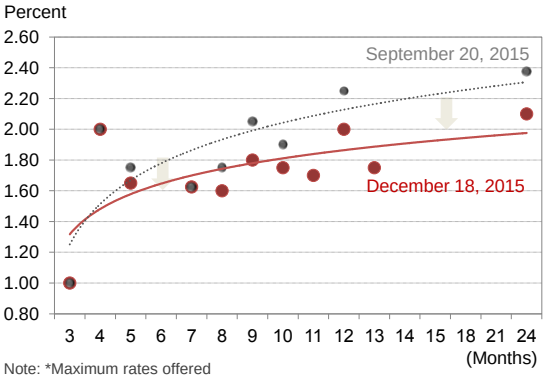
Stability of financial institutions

Overall stability of financial institutions was assured, given the high level of loan loss provision and capital buffers acting as a cushion against deteriorating loan quality.

Loan quality of commercial banks deterriorated as a whole, as the slow and uneven economic recovery affected repayment ability. This was reflected by a rise in the non-performing loan (NPL) ratio to 2.8 at the end of 2015 Q3 from 2.4 at the end of the previous quarter **(Chart 3.22).** The fall in loan quality stemmed from large manufacturing companies affected by the global economic slowdown and depressed commodity prices, as well as from the SMEs hit by the slow domestic economic recovery. Meanwhile, consumer loan quality dropped across every loan category **(Chart 3.23).**

Despite a decline in loan quality, financial system stability remained sound due to prudent risk management by commercial banks. This included sufficiently high levels of loan loss provision and capital buffers. The actual/regulatory loan loss provision ratio stood at 153.9 and the capital adequacy ratio was at 17.3, well above the regulatory requirement of 8.5.

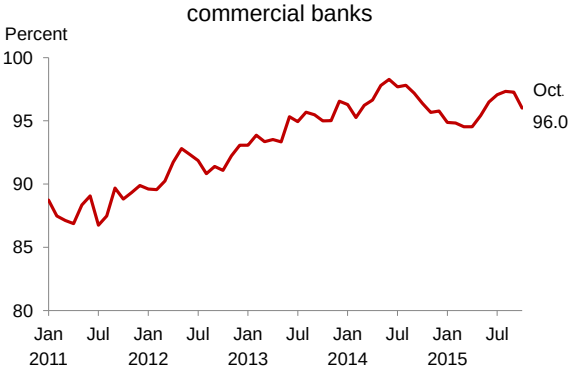
Chart 3.20 Special deposit rates*



Note: *Maximum rates offered

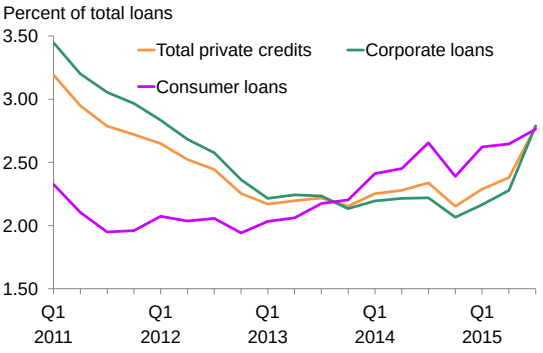
Source: Bank of Thailand

Chart 3.21 Loans to deposits and B/E ratio of commercial banks



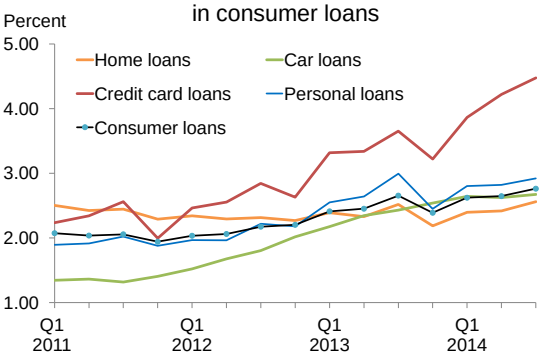
Source: The Bank of Thailand

Chart 3.22 Non-performing loans of commercial banks



Source: Bank of Thailand

Chart 3.23 Share of non-performing loans in consumer loans



Source: Bank of Thailand

3.3 Non-financial sectors

Household sector

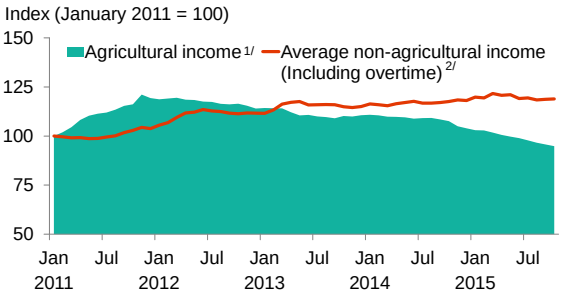
Slow economic recovery and depressed commodity prices led to the reduction in household incomes and debt repayment ability. Farm households were the most fragile.

Sluggish economic recovery dented household incomes. Farm incomes continued to contract given distressed farm production and prices. Meanwhile, non-farm incomes (including overtime) remained stable **(Chart 3.24).** As a result, households moderated spending on durable goods and slowed down additional borrowing, as indicated by the decline in growth of household debt as a share of gross domestic product from 6.5 percent at the end of 2015 Q1 to 6.2 percent at the end of 2015 Q2 **(Chart 3.25).**

Declining incomes, coupled with high indebtedness resulting from accelerated debt accumulation over the recent years, affected loan repayment ability of households—particularly, agricultural households and those with non-agricultural businesses **(Chart 3.26).** For both groups the debt service ratio (DSR) at the 75th percentile, representing households with relatively large debt burden, rose markedly from the levels observed in 2013, more than any other groups, and stood close to the 40 percent[5/] threshold, indicating a high probability of default. This was consistent with a growing non-performing loan (NPL) ratio of commercial banks, which at the end of 2015 Q3 stood at 2.8 percent, up from 2.7 percent in the previous quarter. However,

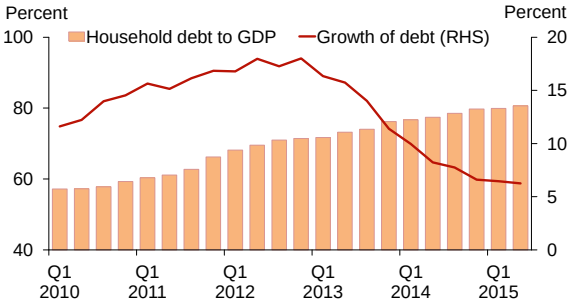
[5/] BOT Symposium 2013, "Rising Household Debt: Implications for Macroeconomic Stablility"

Chart 3.24 Household income



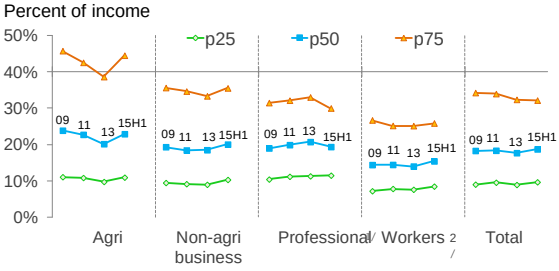
Note: 1/ Seasonally adjusted, 12-month moving average
2/ Seasonally adjusted, 3-month moving average
Source: Office of Agricultural Economics and National Statistical Office, calculations by the Bank of Thailand

Chart 3.25 Household debt[1/] to GDP[2/]



Note: 1/ Loans to households by financial institutions
2/ Computed using fix-based GDP (base year = 1988)
Source: Bank of Thailand

Chart 3.26 Debt Servicing Ratio: DSR



Note : Only households with debt are considered.
1/ Including managers, academics, professional workers, and technical workers
2/ Such as workers in agriculture, forestry, fishery, services, machine operations, as well as clerks and craftsmen in manufacturing and construction.
Source: The household socioeconomic survey, National Statistical Office of Thailand, and calculations by the Bank of Thailand

government policies targeted at households, such as measures to improve the living standards at the village and sub-district levels, and soft loans for farmers, would somewhat contribute to lower expenses and higher liquidity among households.

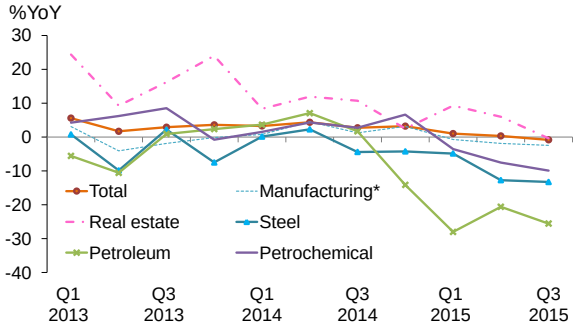
Going forward, the Committee would continue to closely monitor the economic recovery, which would have an important bearing on income and repayment ability of households. Farm households, in particular, remained fragile given the higher debt burden, relative to other households, on account of the draught in the first half of 2016 and depressed global commodity prices.

Corporate sector

Listed corporate earnings in 2015 Q3 softened slightly from the previous quarter, but overall financial health remained sound. Going forward, the economic recovery and effectiveness of government stimulus measures would have to be closely monitored as these would influence corporate revenues and financial standings.

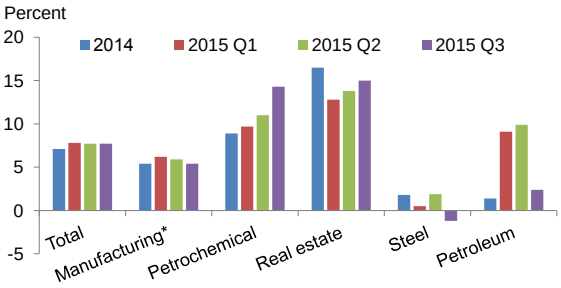
Sales of listed companies (in SET and MAI) in 2015 Q3 declined from the previous quarter due to the slow economic recovery and falling commodity prices **(Chart 3.27).** Moreover, debt repayment ability deteriorated, as suggested by the decrease in the Annualized Interest Coverage Ratio (ICR), especially in the petroleum industry which saw a large fall in sales and a decline in profits due to losses from inventory valuation following plunging global oil prices. Likewise, the decline in iron ore prices led to falling sales, and thus lower profitability, of the iron industry **(Chart 3.28)**. Nevertheless, weakened financial positions, especially in the iron industry,

Chart 3.27 Sales growth classified by business sectors



Note: Median of each business sector
*Excluding petroleum and petrochemical industries
Source: The Stock Exchange of Thailand, calculations by the Bank of Thailand.

Chart 3.28 Operating Profit Margin (OPM) classified by business sectors



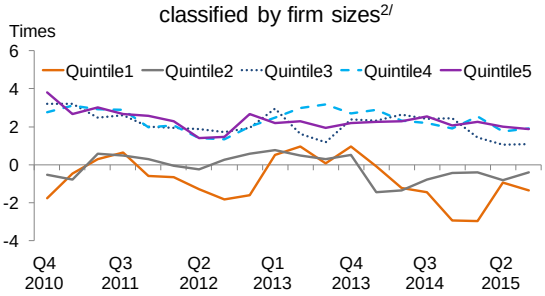
Note: Median of each sector
*Excluding petroleum and petrochemical industries
Source: The Stock Exchange of Thailand, calculations by the Bank of Thailand.

posed limited risks to overall corporate financial stability. This was because the industry contributed to only 2.3 percent of outstanding debt of listed companies, and its median debt to equity ratio (D/E) stabilized around at a low level of 0.6. Meanwhile, profitability of the petroleum industry should recover, as cheaper crude prices would help lower production costs.

If listed companies are divided into five groups according to their asset sizes, the two groups with the lowest amount of assets continued to suffer losses **(Chart 3.29).** If these smaller companies were used to proxy the health of non-listed small and medium firms (SMEs), it may reflect a worsening debt repayment ability of SMEs. Indeed, this was consistent with rising non-performing loan (NPL) ratio among the SMEs **(Chart 3.30).**

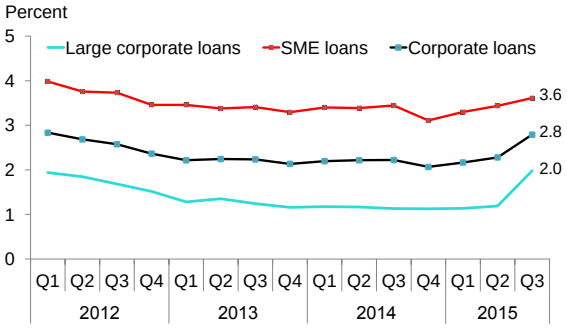
Nevertheless, corporate financial position broadly remained sound. Overall profitability declined only slightly, as indicated by the operating profit margin (OPM) that remained stable from the previous quarter. In addition, profitablity improved in some industries such as petrochemical (which benefited from declining production costs in line with lower crude prices) and the real estate sector (which managed to reduce transaction costs and administration expenses). Regarding debt repayment ability, although ICR declined somewhat, it stood above the international standards of 1.5. In addition, the majority of companies maintained a manageable level of debt and a sufficient amount of liquidity to withstand increasing risks, as reflected by the debt to equity (D/E) ratio and the current ratio (CR) that remained stable from the previous quarter at 0.8 and 1.6, respectively.

Chart 3.29 Interest Coverage Ratio (ICR)[1/] classified by firm sizes[2/]



Note: [1/] 25th percentiles representing companies with low repayment ability. Annualized ICR are shown.
[2/] Companies are classified by asset sizes into 5 quantiles, from smallest to largest.
Source: The Stock Exchange of Thailand, calculations by the Bank of Thailand.

Chart 3.30 NPL ratio of corporate loans classified by firm sizes



Source: Bank of Thailand

Going forward, two key factors would require close monitoring. First, if economic recovery continued to be slow and uneven, then the already fragile sectors, such as SMEs and commodities-related industries, may experience more adverse impacts. Second, conditional on the effectiveness of government stimulus policies, if large-scale investment projects could indeed be completed according to schedule, that would bolster economic growth and boost corporate financial positions.

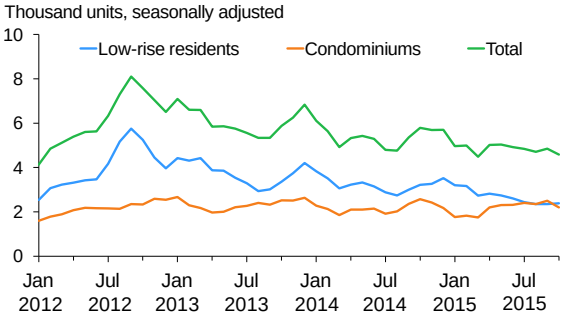
Real estate sector

Both the supply of and demand for real estates in Bangkok and its vicinities slowed down, but government stimulus measures were expected to boost demand and reduce excess supply somewhat.

Demand for real estates in 2015 Q3 slowed down, since purchasing power remained weak and consumers put off purchases awaiting the government's real estate stimulus package that would take effect in late October. Softer demand was also reflected in the decline in loan approvals for residential mortgage especially in the low-rise units **(Chart 3.31),** although the demand for condominiums was steady.

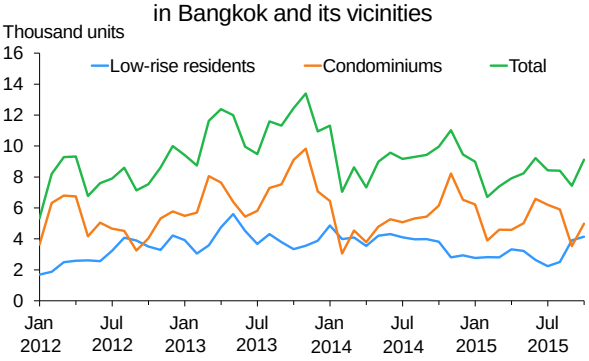
Similarly, the supply of real estate in 2015 Q3 fell from the previous quarter, as indicated by the decline in new residential project launches **(Chart 3.32)**. Low-rise units accounted for the majority of the decline, while the supply of condominiums edged up slightly thanks to the new projects along the extentions of mass transit rail routes. Moreover, launches of high-end residential projects slowed down after a large pickup in the first half of 2015 **(Chart 3.33).** According to surveys of large property developers, the decline in high-end

Chart 3.31 New residential units with loans from commercial banks in Bangkok and vicinity



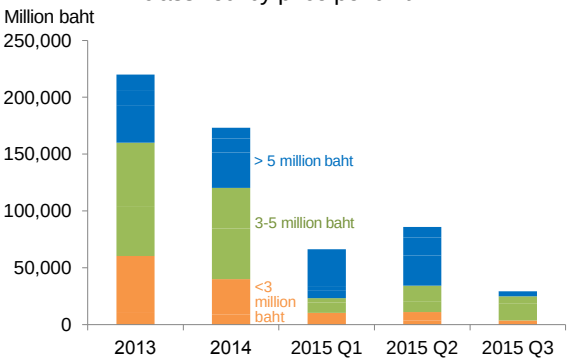
Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA), calculations by the Bank of Thailand

Chart 3.32 New residential project launches in Bangkok and its vicinities



Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA), calculations by the Bank of Thailand

Chart 3.33 Value of new residential projects classified by price per unit



Source: Agency for Real Estate Affairs (AREA)

projects reflected adjustments among developers to weaker demand for such projects. Therefore, this should reduce the risk of excess supply of high-end condominiums.

Going forward, the government's real estate stimulus measures[6/] should incentivize prospective home buyers to expedite purchases and transfers. Moreover, the stimulus measures should help reduce excess supply, particularly of the units priced below 3 million baht that would benefit most from the policy.

Fiscal sector

Stability of the fiscal sector continued to be sound overall. Treasury cash balance remained adequate to buffer the lower-than-projected tax revenue collection. Moreover, the level of public debt remained manageable. In the future, public debt would likely rise due to the new borrowings to cover rising budget deficit, investments in transportation infrastructures, and increasing fiscal liabilities associated with economic stimulus packages.

Fiscal sector stability remained sound. First, treasury cash balance increased and remained sufficient to meet unexpected needs and offset any projected shortfall in revenue collection this year (which would be the second consecutive year[7/]).

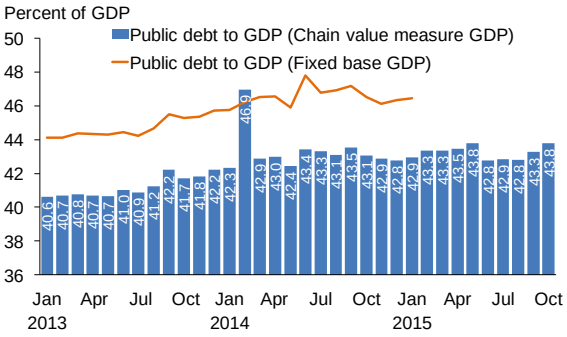
[6/] Real estate stimulus measures include 1) loans not exceeding 3 million baht from Government Housing Bank to support households with low to medium income, 2) reduction in transfer and mortgage fees from 2 percent to 0.1 percent, and 3) first-time homebuyer personal income tax credit.

[7/] Revenue collection shortfall was projected at 118 billion baht, as a result of the worse-than-expected economic slowdown and the decline in tax collection related to the fall in global oil price.

Treasury cash balance at the end of September 2015 stood at 426 billion baht, up from 264 billion baht in June 2015. The increase was due to additional revenue submissions by state-owned enterprises and borrowings to cover budget deficit in the last quarter of fiscal year 2015. Furthermore, revenues from the 4G spectrum auctions on November 11 and December 15 would boost the treasury cash balance. Second, while the public debt to GDP ratio increased slightly, it remained well below the fiscal sustainability threshold of 60 percent. In October 2015, the ratio stood at 43.8 percent, up from 42.8 percent in June 2015 **(Chart 3.34).** Borrowings to cover budget deficit, as well as investments in water resource management projects and the urgent phase of the road transportation development plan (under the second phase of the economic stimulus package) by the central government, largely accounted for the increase in the debt to GDP ratio.

Looking ahead, fiscal stability would be affected by the following factors. First, additional borrowings needed to cover budget deficit would rise to 2.9 percent in fiscal year 2016, up from 1.9 percent of GDP in the previous fiscal year. Second, the government would need to borrow extra-budgetary funds for the construction of three new motorway routes in accordance with the Transportation Action Plan that was pushed back from the previous year. Third, additional financial obligations would likely rise given quasi-fiscal activities as prescribed by the economic stimulus packages approved in fiscal years 2015-2016. These included, for instance, compensations to the Thai Credit Guarantee Corporation for SME loan guarantee fees, compensations to financial institutions taking part in the soft loan programs for the reduced interest incomes, and the government's forgone revenues due to various tax rebates.

Chart 3.34 Public debt to GDP ratio*



Note: (1) Chart shows calendar years, (2) Official figures for Debt-to-GDP ratio based on Chained-Volume Measure GDP are only available for February and March 2015. The figures prior to this period are calculated by the Bank of Thailand

Source: Public Debt Management Office

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014			2015				
		H1	Q3	Q4	Q1	Q2	Q3	Oct	Nov
1. Financial markets sector									
Bond market									
Bond spread (10 years-2 years)	1.3	1.4	1.2	1.4	1.4	1.4	1.2	1.1	1.2
Equity market									
SET Index (End of period)	1,497.7	1,485.8	1,585.7	1,497.7	1,505.9	1,504.6	1,349.0	1,394.9	1,359.7
Actual volatility (SET Index)1/	12.5	14.1	8.4	13.4	12.8	11.3	12.3	11.0	9.8
Price to Earnings Ratio (times)	17.8	17.9	18.5	17.8	20.9	20.1	17.8	18.5	23.8
FX market									
Actual volatility (baht) (%annualized)2/	4.0	4.3	3.8	3.4	3.7	5.4	5.2	7.6	6.2
Nominal Effective Exchange Rate (NEER)	104.3	102.7	104.6	107.0	111.7	109.9	106.6	105.4	106.3
Real Effective Exchange Rate (REER)	103.0	101.9	103.4	105.0	108.0	106.0	102.4	101.3	N.A.
2. Financial institutions sector3/									
Minimum lending rate (MLR)4/	6.8	6.8	6.8	6.8	6.6	6.5	6.5	6.5	6.5
12-month fixed deposit rate4/	1.7	1.7	1.7	1.7	1.5	1.5	1.5	1.4	1.4
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.8	15.9	17.1	16.8	16.6	16.8	17.3	N.A.	N.A.
Earnings and profitability									
Net profit (billion baht)	214.1	111.6	53.9	48.7	50.4	53.3	50.1	N.A.	N.A.
Return on assets (ROA)	1.2	1.5	1.3	1.2	1.2	1.3	1.2	N.A.	N.A.
Liquidity									
Loan to deposit and B/E	95.7	98.3	97.2	95.7	94.5	96.5	97.3	N.A.	N.A.
3. Household sector									
Household debt to GDP (times)	79.7	77.4	78.5	79.7	79.9	80.6	N.A.	N.A.	N.A.
Financial assets to debt (tim	2.5	2.5	2.5	2.5	2.6	2.5	N.A.	N.A.	N.A.
NPL and delinquency ratio (%)									
Thai commercial banks :									
Consumer loans	5.9	5.9	6.2	5.9	6.1	6.2	6.3	N.A.	N.A.
Housing loans	3.8	3.8	4.1	3.8	4.1	4.0	4.4	N.A.	N.A.
Auto leasing	10.8	10.5	10.7	10.8	10.9	11.1	11.2	N.A.	N.A.
Credit cards	5.4	5.5	6.0	5.4	6.3	6.8	7.1	N.A.	N.A.
Other personal loans	5.0	5.0	5.6	5.0	5.4	5.3	5.4	N.A.	N.A.
4. Non-financial corporate sector 5/									
Operating profit margin (%)	7.1	7.1	7.4	6.9	7.8	7.7	7.7	N.A.	N.A.
Debt to equity ratio (times)	0.8	0.8	0.8	0.8	0.7	0.7	0.8	N.A.	N.A.
Income coverage ratio (times)	6.0	6.2	5.9	5.6	5.3	5.2	5	N.A.	N.A.
Current ratio (times)	1.6	1.6	1.6	1.6	1.7	1.6	1.6	N.A.	N.A.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014			2015				
		H1	Q3	Q4	Q1	Q2	Q3	Oct	Nov
4. Non-financial corporate sector 5/ (Continues)									
Delinquency and NPL ratio									
Large bussiness	1.1	1.2	1.1	1.1	1.1	1.2	2.0	N.A.	N.A.
SMEs	3.1	3.4	3.4	3.1	3.3	3.4	3.6	N.A.	N.A.
5. Real estate sector									
The number of approved mortgages from banks (Bangkok and its vicinity)	62,839	16,315	17,345	16,299	11,564	14,939	15,770	3,834	N.A.
Single-detached and semi-detached houses	15,694	4,206	4,230	3,927	3,001	3,383	3,115	721	N.A.
Townhouses and commercial buildings	21,764	5,921	5,844	5,215	4,212	4,987	4,812	1,282	N.A.
Condominiums	25,381	6,188	7,271	7,157	4,351	6,569	7,843	1,831	N.A.
The number of new openings (Bangkok and its vicinity)	111,211	28,714	28,268	28,367	22,191	27,677	22,283	10,970	N.A.
Single-detached and semi-detached houses	18,933	5,657	5,349	3,173	3,231	2,045	6,694	1,803	N.A.
Townhouses and commercial buildings	26,980	7,261	6,611	5,609	5,172	5,866	5,014	944	N.A.
Condominiums	65,298	15,796	16,308	19,585	13,788	19,766	10,575	8,223	N.A.
Housing price index 6/									
Single-detached houses (including land)	117	115.5	119.3	118.4	119.4	118.4	122.0	122.0	N.A.
Townhouses (including land)	130.2	128.6	134.2	132.5	135.8	138.2	142.0	141.1	N.A.
Condominiums	154.3	153.1	156	161.9	162.9	164.4	171.6	177.8	N.A.
Land	150.7	148.3	153	155.5	163.1	163.3	169.5	171.5	N.A.
6. Fiscal sector									
Public debt to GDP (%)	46.3	47.8	47.2	46.3	43.3	42.8	43.0	N.A.	N.A.

1/ Annualized standard deviation of return

2/ Daily volatility (using exponentially weighted moving average method)

3/ Based on data of all commercial banks

4/ Average value of 4 largest Thai commercial banks

5/ Only listed companies on SET (median); with revisions to data and methodology.

6/ Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change